Rediscover the walking symphony with Sildenafil

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INDICATION(S) (TARGET)	Peripheral arterial disease with intermittent claudication
PRINCIPAL INVESTIGATOR	Dr. Loukman Omarjee, Ding, MD, MSc, PhD Vascular Physician and Clinical Immunologist NuMeCan Institute – EXPRES Team – UMR Inserm U1241 & CIC-1414
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SIGNATURES

INVESTIGATOR'S SIGNATURE

I have read all pages of this clinical trial protocol for which the CHU of Rennes is the sponsor. I confirm that it contains all information necessary for the correct conduct of the study. I agree to conduct the study in compliance with the protocol and terms of conditions which are defined in it. I agree to conduct the trial in accordance with the applicable laws, regulations and good clinical practice.

I also agree that investigators and other qualified members of my team may have access to copies of this protocol and documents pertaining to the correct conduct of the study enabling them to work in compliance with conditions contained in these documents.

NAME: Dr. Loukman OMARJEE

Signature: Date:

SPONSOR'S SIGNATURE

Sponsor: CHU de Rennes

NAME: Director of Research

Signature: Date:

LIST OF ABBREVIATIONS

ABI	Ankle-brachial index
ACD	Absolute Claudication Distance
ACE	Angiotensin Convertase Enzyme
AE	Adverse Event
ANSM	Agence Nationale de Sécurité du Médicament et des produits de santé (French Health Products Safety Agency)
AT2	Angiotensin 2
CHF	Congestive Heart Failure
CHU	Centre Hospitalier Universitaire (University Hospital)
CINC-1	Cytokine-Induced Neutrophil Chemoattractant-1
CRA	Clinical Research Assistant
CPP	Comité de Protection des Personnes (French Ethics Committee)
CNIL	Commission Nationale de l'Informatique et des Libertés (French Data Protection Agency)
CRF	Case Report Form
CRT	Clinical research technician
CMBP	Central Mean Blood Pressure
CSBP	Central Systolic Blood Pressure
Cwk	Arterial compliance
DBP	Diastolic Blood Pressure
DCI	International Common Denomination
DLCO	Diffusion capacity of the Lungs for carbon monoxide (CO)
DRLQ	Disease-related quality of life
DROP	Decrease from rest of oxygen pressure
DSMB	Data and Safety Monitoring Board
ECQ	Edinburgh claudication questionnaire
ED	Erectile dysfunction
EDHF	Endothelium-Derived Hyperpolarizing Factor
EFS	Event Free Survival
cGMP	Cyclic Guanosine Monophosphate
FDA	Food and Drug Administration
FMD	Flow-mediated vasodilation

HAS	Haute Autorité de Santé
IC	Intermittent Claudication
ICH	International Conference on Harmonisation (Conférence internationale pour l'harmonisation)
IIEF-5	'International Index of Erectile Function' questionnaire
IL-1	Interleukin 1
IMT	Intima-media thickness
INSERM	Institut National de la Santé et de la Recherche Médicale (National institute for medical research)
LC/MS	Liquid Chromatography/Mass Spectrometry
LSCI	Laser Speckle Contrast Imaging
MACE	Major Adverse Cardiovascular Events
MAP	Mean Arterial Pressure
MWD	Maximum walking Distance
MWT	Maximum walking time
NADPH	Nicotinamide Adenine Dinucleotide Phosphate-oxidase
NAION	Non-arterial ischemic Ophtalmic Neuropathy
NO	Nitric Oxide
OR	Odds ratio
OS	Overall survival (used in the expression OS rate)
PDEi	Phosphodiesterase inhibitor
PDE5i	Phosphodiesterase type 5 PDE inhibitor
PAD	Peripheral Arterial Disease
PAQ	Peripheral Artery Questionnaire
PFWT	Pain-Free Walking Time
PORH	Post Occlusive Reactive Hyperemia
PP	Pulse Pressure
PPRIGO	Production Pharmaceutique pour la Recherche Institutionnelle du Grand Ouest
PWV	Pulse wave velocity
QoL	Quality of Life
QSP	Sufficient quantity for (used to define the drug content)
SBP	Systolic Blood Pressure
SAE	Serious Adverse Event
SF-36	"Short-form Health-Related Quality of life" questionnaire

SmPC	Summary of Product characteristics
SSI	Surgical Site Infection
TBI	Toe to Brachial Index
TcpO2	Transcutaneous oxygen pressure
TPR	Vascular resistance
TNF	Tumour Necrosis Factor
TT	Transit Time
UMR	Unité mixte de recherche (Research team)
US	United States (of America)
WD	Walking Distance
WELCH	"Walking estimated limitation calculated by history" questionnaire.
°C	Degree Celsius

1 GENERAL INFORMATION

1.1 Title

VIRTUOSE – Efficiency of sildenafil on the absolute claudication distance of peripheral arterial diseased patients with intermittent claudication. A phase III, National, Multicenter, Prospective, Randomized, Double-blind, Placebo-controlled trial.

1.2 Sponsor

1.2.1 Identification

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Rennes University

1.4.2 Associated investigators

See appendix « List of principal investigators »

1.5 Associated scientific partners

See appendix « Study Team »

1.6 Pharmacy

See appendix « Study Team »

1.7 Pharmacovigilance

See appendix « Study Team »

1.8 Methodologist – biostatistician

See appendix « Study Team »

1.9 Scientific Committee

The mission of the Scientific Committee is to settle scientific, methodological and ethical problems raised by the trial and to monitor its proper conduct. It will be constituted at the beginning of the study with 5 members, including the coordinating investigator, the methodologist, a representative from the sponsor, a researcher in vascular biology and one member among the associated teams.

The Scientific Committee will solve practical problems arising during the course of study and may decide to amend the protocol. If such amendments are done, the relevant Independent Ethics Committee and the Competent Authority will be notified. The Scientific Committee will send to investigators all the information necessary for the proper conduct of the study.

The Scientific Committee composition is described in the appendix.

1.10 Independent Data and Safety Monitoring Board

An independent Data and Safety Monitoring Board (DSMB) is not needed for this study because:

- The investigational medicinal product (sildenafil) has already been authorised on the European market since 1999 (Viagra®), already indicated in elderly men, and its side effects are well known.
- The patient selection criteria take into account contraindications.
- No interim analysis is planned for this study.

2 SCIENTIFIC RATIONALE AND GENERAL DESCRIPTION OF STUDY

2.1 Epidemiology of peripheral arterial disease (PAD)

The worldwide prevalence of lower extremity PAD is estimated at about 15% of elderly patients[1–3]. In the United states (U.S.) National Health and Nutrition Examination Survey, the prevalence of PAD was 0.9% in patients aged 40 to 49 years, and increased to 14.5% in patients aged > 69 years[4]. The prevalence of PAD is expected to increase worldwide[3] with a conservative estimate of >202 million and about 7 million people in France afflicted with this disease globally and a relative increase in PAD prevalence of 23.5% during the first decade of the new millennium[1]. In 2001, the U.S. Medicare program spent an estimated >$4.3 billion on PAD related treatments[3]. PAD-related treatment accounted for 13% of all Medicare Part A and B expenditures for patients undergoing treatment for PAD and for 2.3% of total Medicare Part A and B expenditures. These Medicare costs have continued to increase with an estimated total cost of vascular-related hospitalisations of $21 billion in the United States in 2004[5]. Multiple studies have established that patients with Intermittent Claudication (IC) are at increased risk of cardiovascular events, whereas the risk of major amputation is very low (<1% per year)[2,6]. IC is generally the first clinical sign or symptom of PAD[7]. IC is defined as a reproducible discomfort and/or pain in the lower limbs that is induced by exercise and then relieved with rest. IC is caused by arterial obstruction proximal to the affected muscle bed, thereby attenuating exercise-induced augmentation of blood flow leading to transient muscle ischemia[8–10]. Claudication severely affects Quality of Life (QoL). IC is associated with severe functional impairment that can be significantly improved by intervention in carefully selected patients.

Multiple studies have objectively documented the adverse effect of PAD and claudication on patients' functional status[11–15]. Although claudication patients have a limited progression to chronic limb ischemia, PAD is associated with increased risk of cardiovascular morbidity and mortality. Patients in the lowest quartile of an office-administered 6-minute walk performance test exhibited significantly increased mortality (odds ratio [OR]=2.36+IC95%).
This fact, along with the impact on lifestyle and the large population affected, has caused PAD to be identified as an area where new therapies are urgently needed[15].

2.2 Assessment of functional impact of peripheral arterial disease (PAD)

Originally, the most referenced classification was that of Leriche and Fontaine which defined the stage 2a as mild claudication and the stage 2b as moderate to severe claudication. To date, the most common classification used is the classification of Rutherford that characterizes mild, moderate to severe claudication as the stage I categories 1, 2 and 3. The classification of Rutherford defines mild claudication as the ability to walk >200m, moderate claudication as the ability to walk 100-200m and severe claudication as the ability to walk <100m. The classification of Rutherford defines also claudication as the inability to walk 5 minutes (267m) during a standardised treadmill walking test (speed 2 miles/h – 3.2 km/h; grade 12%). In the clinical setting, the protocol most commonly used is 3.2 km/h (~2 miles/h) with 10% grade instead of 12% because this slope is often too difficult for elderly patients. Please note that although rarely mentioned in papers citing the classification, this treadmill duration clearly appears in the original paper[16].

2.3 Management of PAD

According to recommendations, establishing an appropriate medical treatment and lifestyle modification for a minimum of 3 to 6 months[17], should always precede consideration for invasive procedures. Indeed, most PAD patients suffer a vicious circle evolution leading to approximately 30% of the patients with claudication undergoing revascularization within the 6 months follow-up period, approximately 5% dying per year and a further 5% presenting cardio-vascular complications per year[18–20]. This point is very important in our study because it justifies the fact that an evaluation of the patients at month 6 does not represent a loss of chance for the patients. It is also important to keep in mind that patients in the VIRTUOSE study are not asymptomatic patients or patients with minor claudication but are patients with moderate to severe claudication despite treatment that are yet potential candidates for surgery.

2.3.1 Recommended treatment for patients with IC

Patients with IC carry a significant systemic burden of atherosclerosis and are at risk from the associated complications. These patients should have lifelong treatment designed to reduce atherosclerotic risk factors. Recommendations from the European Society of Cardiology on the diagnosis and treatment of PAD[17] advise the following approaches:

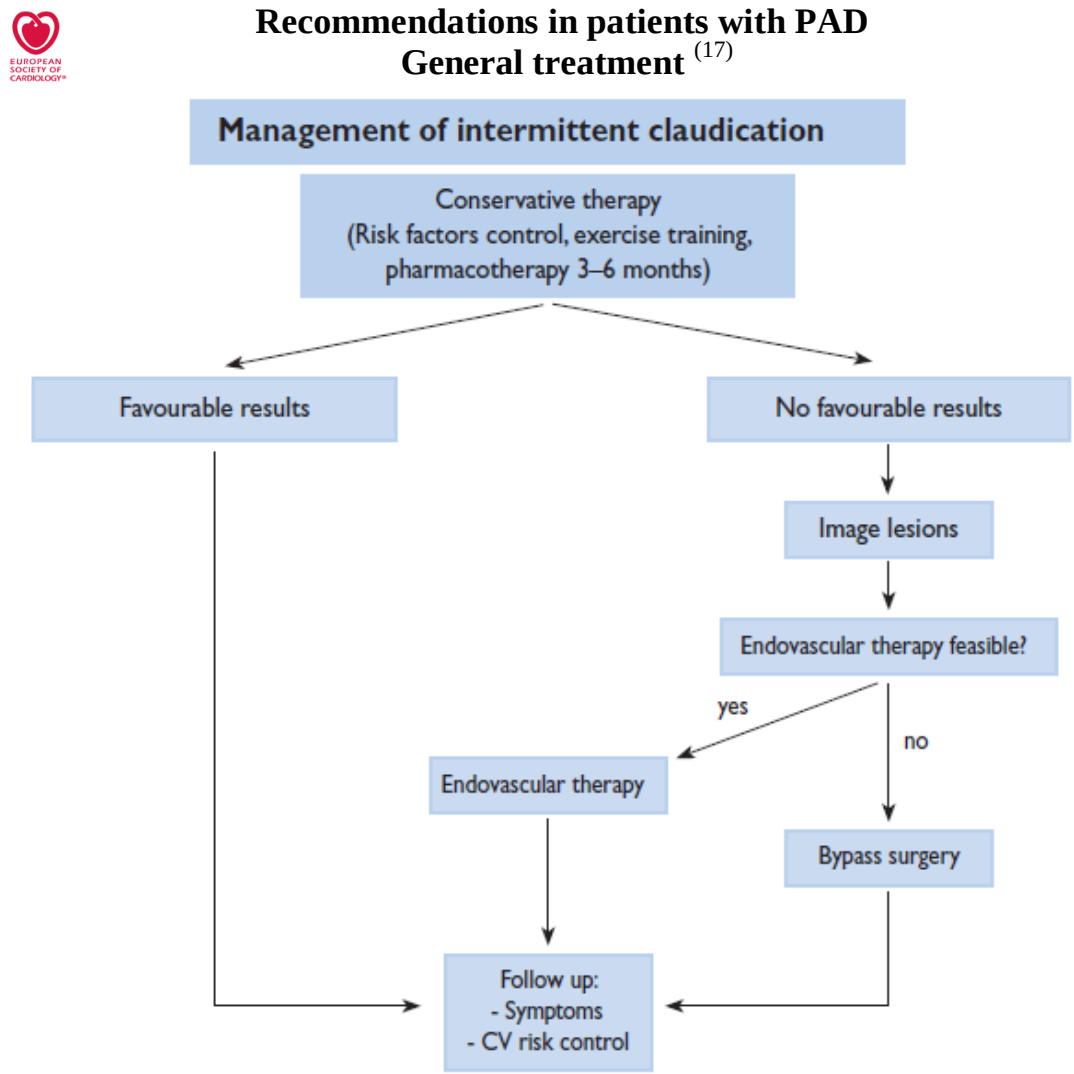


Figure 1: ESC Guidelines for Management of PAD[17]

According to the French Recommendations from the "Haute Autorité de Santé" (HAS) about the management of peripheral arterial disease, the optimal medical treatment must include ACE inhibitor / AT2 antagonists, Lipid Lowering Drugs and Antiplatelet agents unless contra-indicated.

2.3.2 Control of the risk factors

Risk factors include smoking, diabetes, hypertension and other modifiable risk factors[11,21–28].

2.3.3 Lipid Lowering Drugs

Treatment of dyslipidaemia with statins reduces the likelihood of adverse cardiovascular events in patients with atherosclerosis[29–32]. Statin therapy has also improved pain-free walking time in small studies of patients with IC[33,34], although the results are controversial[35].

2.3.4 Antiplatelet agents

Numerous studies have demonstrated the benefit of antiplatelet therapy, especially aspirin, in doses of 75 to 325 mg/d in reducing rates of myocardial infarction, stroke, and vascular-related deaths in patients with symptomatic lower extremity atherosclerosis[36]. The American Heart Association guidelines for lower extremity ischemia rated this treatment as recommendation class I-A[37]. In the 6,452 patients with PAD in the Clopidogrel vs Aspirin In Patients At Risk of Ischemic Events (CAPRIE) trial, Clopidogrel reduced the myocardial infarction, stroke, or vascular death rate by 23.8% more than aspirin alone[38]. Although a single study demonstrated that combination aspirin and Clopidogrel therapy was associated with a 20% relative risk reduction for myocardial infarction, cardiovascular death, or stroke[39], there is no evidence to date that combination therapy is a more effective treatment for PAD than a single agent, and bleeding risks are increased[40].

2.3.5 Anticoagulant agents

In the initial version of the protocol, the presence of an antiplatelet agent is mentioned in the optimal medical treatment. However, many patients may be on direct oral anticoagulants (Rivaroxaban, Apixaban, Dabigatran) to prevent the risk of thrombosis related to atrial fibrillation (AF) or as part of the acute treatment of venous thromboembolism (deep vein thrombosis, pulmonary embolism) or as a long-term treatment to prevent the risk of recurrence[41,42]. In addition, it has been shown that in PAD patients, there was an increased risk of AF, ischemic stroke, and mortality[43].

New data from the COMPASS study[44] show that treatment with Rivaroxaban 5mg bid is non-inferior to treatment with Aspirin 100mg daily on the incidence of major adverse cardiovascular events (MACE) (cardiovascular mortality, myocardial infarction, stroke), all-cause mortality with a higher risk of bleeding than treatment with Aspirin alone[45].

If prolonged anticoagulant therapy is required in PAD patients, the European Society of Cardiology on the diagnosis and treatment of PAD recommend this therapy as an alternative to antiaggregant therapy[46].

However, based on the results of the WAVE trial, the American Heart Association (grade IIIA) and the European Society of Vascular Medicine (ESVM) recommend against the use of a vitamin K antagonist (VKA) to reduce the risk of MACE in PAD patients[47].

To remain in line with the reality of the field and considering the above argument, PAD patients with claudication could be included if they are under optimal treatment for one month including direct oral anticoagulant (Rivaroxaban, Apixaban, Dabigatran) as monotherapy instead of an antiplatelet agent if the treatment is justified for another cause than MI (non-valvular AF, VTE).

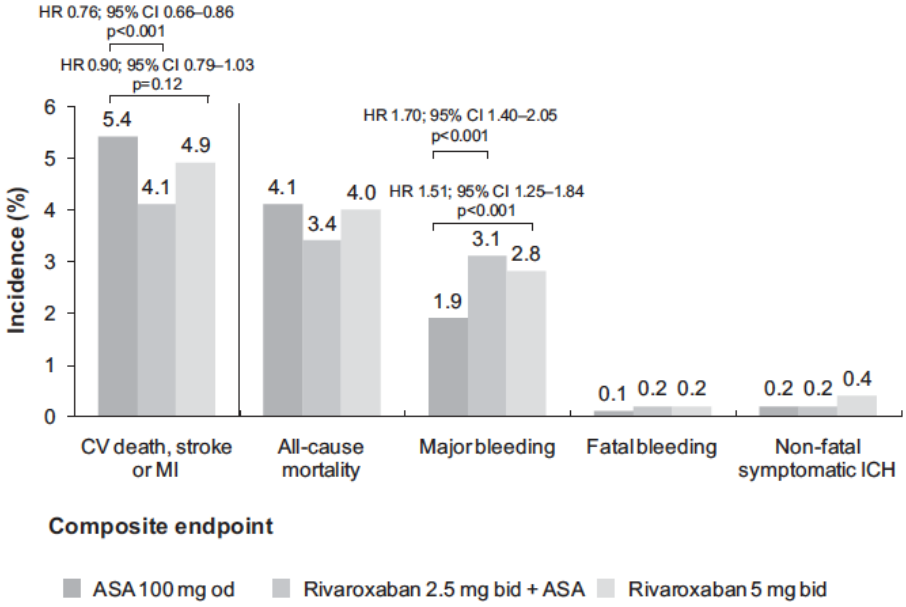


Figure 2: Clinical outcomes from COMPASS primary analysis[44]. ASA: acetylsalicylic acid; CI: Confidence Interval; CV: Cardiovascular; HR: Hasard ratio; ICH: Intracranial bleeding; MI: Myocardial Infarction; od: once daily

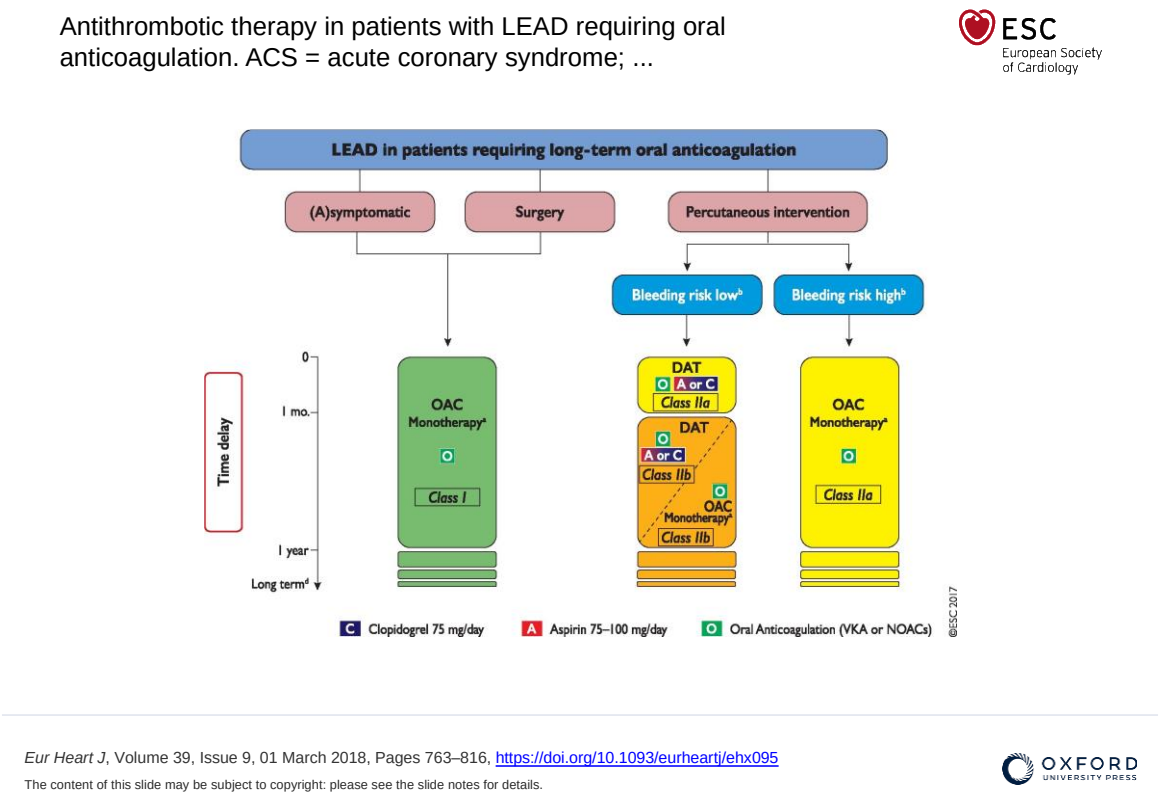


Figure 3: Antithrombotic therapy in PAD patients requiring oral anticoagulation[46].

2.3.6 ACE inhibitors and AT2 antagonists

The ACE inhibitor Ramipril has beneficial effects in patients with PAD and IC. In the Heart Outcomes Protection Evaluation study[48], treatment with Ramipril reduced cardiovascular events and mortality even in patients without hypertension. Therefore, Ramipril should be considered as a first-line choice for hypertension treatment in PAD patients, although it should be used with caution in the presence of renal artery stenosis. In a recent double blind, placebo-controlled RCT, Ramipril (10 mg/d for 24 weeks) was associated with a significant improvement in pain-free and maximum treadmill walking times and in measurements of physical function[48].

The AT2 antagonist Telmisartan has beneficial effect in patients with PAD on absolute walking distance and endothelial function[49]. In a single centre, single-blinded, prospective study, 36 patients with PAD at stage Fontaine II or higher and mild to moderate arterial hypertension were treated with telmisartan 40/80 mg once daily or placebo for 12 months. Primary endpoint was the improvement of the absolute treadmill walking distance (WD). Flow-mediated vasodilation (FMD), carotid intima-media thickness (IMT), ankle-brachial index (ABI) and

disease-related quality of life (DRQL) were examined as well. After 12 months, maximum WD increased by 26% in the telmisartan group (P < 0.001). However, in the placebo group it was comparable to baseline. FMD rose by 40% in the telmisartan group while it deteriorated in the placebo group (P < 0.001). IMT and ABI were comparable in both groups at baseline and did not change considerably after 12 months. In non-diabetic patients (72.2%), the ABI did not change in the placebo group, whereas it increased by 11% in the telmisartan group (P < 0.001). While the DRQL remained stable in the telmisartan group, placebo treatment was associated with a marked deterioration (P < 0.01)[49].

Telmisartan improves WD and endothelial function, the ABI in non-diabetic patients and it may prevent further loss of quality of life in patients with advanced PAD[49].

2.3.7 Physical activity

Exercise therapy is a cornerstone in the management of IC and has been the subject of multiple case series, randomized trials, and meta-analyses. Exercise programs for patients with IC have been found to increase the distance to onset of claudication and increase the distance to maximum claudication pain. A meta-analysis of 1,200 patients demonstrated that exercise therapy, compared with placebo or usual care, provided an overall improvement in walking ability from 50% to 200%, with these improvements maintained for up to 2 years[50]. The American Heart Association for many years has considered the quality of the evidence supporting exercise therapy in the treatment of IC to be sufficiently robust to justify a Level I recommendation[37]. Unfortunately, exercise induces pain resulting in a vicious circle (Pain > inactivity > disease progression > pain > increased morbi-mortality). Drugs that could facilitate the return to activity by improving walking capacity may help patients move from this vicious circle to a virtuous circle (no Pain or improved ability for activity > collateral vessel development > slowing of disease progression > decreased morbi-mortality).

2.3.8 Pharmacotherapy for patients with IC to improve walking ability

a) General concept

Medical management of IC is aimed at symptom relief and slowing the progression of atherosclerotic disease. A number of drugs have been evaluated for use in patients with IC, but in the United States, there are currently only two Food and Drug Administration (FDA) approved medications: Cilostazol and Pentoxifylline[51–53]. Of note, a recent review by the Royal College of Physicians in the United Kingdom identified that Naftidrofuryl, widely available in Europe but not FDA-approved in the United States, was the drug of choice over both Cilostazol

and Pentoxifylline in the medical management of symptomatic PAD[54,55]. The idea for these treatments is not necessarily to propose a life-long treatment, but to help patients move from the vicious to the virtuous circle.

b) Pentoxifylline and Naftidrofuryl

Pentoxifylline was the first drug approved by the FDA for IC in 1984 and has been largely studied[51,52,56,57]. The French health authority committee (ANSM) highlights an insufficient efficacy regarding medical improvement. The treatment of PAD with the present drug regarding the highlights of the French health authority committee is that it may be used to treat symptomatic PAD. According to this committee, studies are not of good quality and none of them compared analgesic drugs to the reference analgesic drug which is Paracetamol. Moreover, the impact on the morbi-mortality of Pentoxifylline has never been assessed with PAD patients and the ratio regarding efficiency/risks is unknown. To date, the literature does not provide any evidence concerning the effects of vasodilator drugs on the systemic complications of atherosclerosis or their ability to act in a potentially preventative nature. Regarding the low efficiency for treating leg pain in patients with claudication, this treatment has not been reimbursed by the national health insurance in France since the 1st of March 2012. Naftidrofuryl oxalate may be used in Europe, but it is not currently approved in the United States. It has been shown to increase pain-free walking distance[54–56], nevertheless, the security health survey committee has confirmed potential severe but rare risks on liver function as well as a risk of ulceration of the oral mucosa in treated patients.

c) Cilostazol

Cilostazol is a Phosphodiesterase inhibitor (PDEi) type 3 that suppresses platelet aggregation and is a direct vasodilator. Patients can notice improvement in maximal and pain-free walking distance in delays as short as 4 weeks[53]. Many studies have confirmed these results with short term or long term treatments[53,57], with a satisfactory safety profile[58]. However, Cilostazol has scarcely been used in France due to its cost and its frequent side effects, and it was withdrawn in 2010[59].

2.4 Study treatment

Sildenafil is a phosphodiesterase type 5 inhibitor (PDE5i) that acts by amplifying NO effects via an elevation in the available pool of cyclic guanosine monophosphate (cGMP)[60]. This drug

has been successfully used to treat a number of conditions, including erectile dysfunction, pulmonary hypertension, and congestive heart failure (CHF)[61,62], but to date only a few results are available for IC[63,64].

2.5 Rationale for the use of study treatment for Intermittent Claudication

2.5.1 General comments on PDE5 inhibitors

Available PDE5i include sildenafil citrate, tadalafil, vardenafil, and avanafil. Sildenafil is the first marketed PDE5i and has been extensively studied in the literature. These drugs are oral therapies that are taken before sexual activity[65] and have been extensively evaluated in clinical trials, are efficacious and are well tolerated. They facilitate erection by preventing the breakdown of cGMP and keeping blood vessels dilated for longer[66]. However, their action is not limited to the penile tissue, so that systemic effects could be hypothesised. It has been suggested that chronic administration of PDE5i may improve endothelial function[67], a key mechanism in atherosclerosis and PAD.

Proposed mechanisms of action of PDE5i in improving endothelial function[68–73], include:

- Enhancement of the NO/cGMP signalling cascade[74,75]
- Inhibition of NADPH oxidase activity[76–78]
- Reduction of superoxide formation[67,76,78]

Thus, it seems that PDE5i could restore the physiological equilibrium between mediators of vasoconstriction and vasodilation, inhibiting the catabolism of cGMP and enhancing NO activity and, consequently, vasodilatation[78]. These effects could result in improvement of endothelial function, reducing atherogenesis and vascular complications[78].

2.5.2 PDE5i and ischemia

Animal models have shown positive results on angiogenesis and ischemia for PDE5i[79–84]. Overall, these animal studies appear to provide evidence that PDE5 inhibitors improve survival of skin flaps, angiogenesis, and tissue tolerance to ischemia.

In humans, following a case report[64] and evidence in Raynaud syndrome with regard to severe critical ischemia[84–89], there is only one study available in the literature evaluating the effect of sildenafil on skeletal muscle oxygenation during exercise in men with IC[63]. This randomized, double-blinded, crossover study was conducted in 12 men with assumed stable IC. They received two consecutive doses of 50 mg of sildenafil or matching placebo and underwent a symptom-limited exercise test on the treadmill. Acute administration of sildenafil improved

skeletal muscle oxygenation during exercise but failed to enhance walking tolerance in these patients[63].

Three major limitations arise from this study. First the number of subjects was very small. Second a dramatic increase in walking capacity was found in the placebo group which is clearly not expected in PAD patients except from experience gained using the treadmill. Three the study focused only on an acute effect of the drug.

2.5.3 PDE5i and pain

Another original pathway that may lead to a positive effect on IC in human is the role of PDE5i on the activation of pain through the cGMP / NO pathway:

Alves et al.[90] showed that the L-arginine/NO/cGMP pathway functions as an endogenous modulator of peripheral inflammatory hyperalgesia. Also, *Lee et al.*[91] showed that sildenafil was an effective treatment for acute pain at the spinal level in the induced pain-state model. Additionally, spinal adenosine A1, A2A, A2B, and A3 receptors may play a role in sildenafil-induced anti-nociception. *Rocha et al.*[92] showed that tadalafil, another PDE5i, dose-dependently inhibited hypernociception in zymosan and osteoarthritis models. In zymosan-induced arthritis, tadalafil significantly decreased neutrophil influx and TNF-alpha release but did not alter IL-1 or cytokine-induced neutrophil chemoattractant-1 (CINC-1) levels. Therapeutic oral administration of tadalafil provided analgesia mediated by guanylyl cyclase and was independent of the release of endogenous opioids. This effect of tadalafil was associated with a decrease in neutrophil influx and TNF-alpha release in inflamed joints.

2.5.4 PDE5i and arterial stiffness

Sildenafil has a favourable effect on aortic stiffness and wave reflection in patients with coronary artery disease and the effect lasted throughout the study (3 h), being evident 30 min after drug intake[93]. Also, in patients with heart failure, sildenafil improves cardiac performance because of a decrease in left ventricular load (LV), which is caused by decreases in peripheral resistance, in aortic and large artery stiffness, and in wave reflection from peripheral sites. This can explain the increase in cardiac output and in exercise capacity with sildenafil in patients with heart failure[94]. In patients with erectile dysfunction, In patients with erectile dysfunction who were administered sildenafil, the pulse wave velocity, along with blood pressure, tended to decrease transiently after dosing. There is a possibility that sildenafil affects the improvement of erectile dysfunction via the decrease of pulse wave velocity, especially in patients with risk

factors for arteriosclerosis[95]. In patients with hypertension, sildenafil may reduce arterial stiffness in the recovery period after exercise[96].

2.5.5 Preliminary experience

We have treated, after advice from the ethics committee and as a compassionate treatment, 5 patients with non-revascularizable PAD. The average walking distance before treatment was 259 ± 143 meters and was improved for all patients (Figure 2): on average it increased to 905 ± 374 meters after treatment. (Note that the patient with the lowest improvement was tested on treadmill after only one day of treatment whereas others had 3 months of treatment).

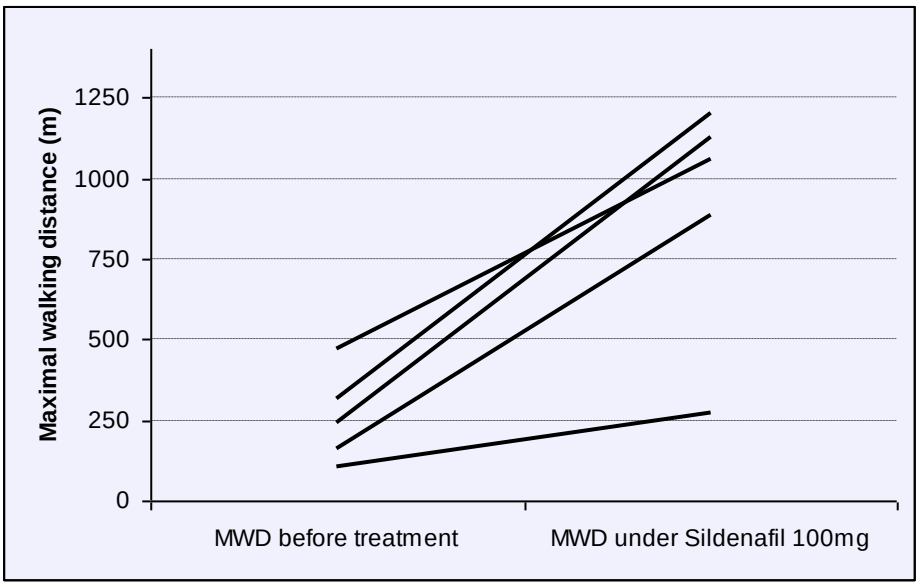


Figure 4: Improvement in Maximal Walking Distance (MWD) of 5 patients with 100 mg sildenafil.

Following is one of these patients, reported as a case study[97]:

This is a 61-year-old patient who had an aorto-bifemoral bypass in 2003. Since 2003, and over a period of 9 years with optimal medical treatment, regular supervised rehabilitation program, he has had a buttock claudication with erectile dysfunction and was repeatedly considered to have non-revascularizable ischemia. He had an impairment of his quality of life that led to his divorce. His walking distance has never exceeded 300 meters since the initial bypass. In February 2015, his walking distance was low, as usual. With an initial dose of 50mg sildenafil, his walking distance increased from 167 meters to 378 meters in a month, but he still had a history of pain during exercise. After increasing the dose to 100 mg per day, his walking distance increased in a week from 378 meters to 713 meters, he had no more pain nor buttock claudication. A few weeks later his walking distance increased to 883 meters still with no pain

(Figure 3). Since then, the patient has been satisfied with the treatment combining optimal medical treatment + sildenafil. He has been regaining confidence and has been resuming physical activity which enabled him to improve his metabolic parameters and ultimately change from the vicious circle of claudication to the virtuous circle of benefits, improving his physical activity and his quality of life, particularly his cardiovascular health[97].

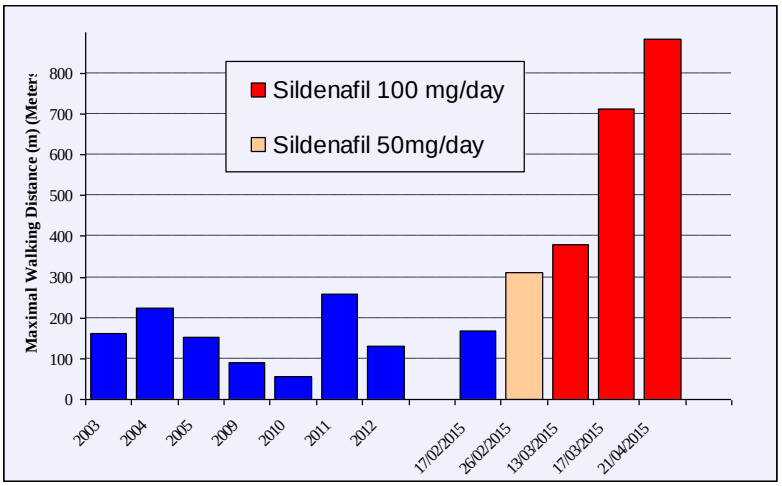


Figure 5: Incredible improvement of Walking Distance Capacity of a patient with proximo distal limb ischemia.

2.5.6 Positive Results from ARTERIOFIL Study

BACKGROUND:

Patients with lower extremity PAD frequently experience claudication, a clinical symptom indicative of reduced walking capacity. Recommended care consists of exercise rehabilitation combined with optimal medical treatment and surgery. The effects of a single oral dose of sildenafil, a phosphodiesterase type-5 inhibitor, on patients with claudication are discussed. The aim of this study was to test the efficacy of a single 100 mg dose of sildenafil compared to placebo in terms of maximal walking time (MWT) in patients with claudication.

METHODS:

The ARTERIOFIL study is a crossover, double-blind, prospective, randomized, single-center study conducted at Angers University Hospital in France. MWT (primary endpoint) was assessed using a treadmill test (10% incline: 3.2 km/h). Secondary endpoints (pain-free walking time (PFWT), transcutaneous oximetry during exercise and redox cycle parameters and safety) were also studied.

RESULTS:

Fourteen patients were included of whom two were ultimately excluded. In the 12 remaining patients, the MWT was significantly improved during the sildenafil period compared with the placebo period (300 s [95% CI 172 s-428 s] vs 402 s [95% CI 274 s-529 s] $p < 0.01$). Sildenafil had no significant effect on pain-free walking time or skin tissue oxygenation during exercise. According to redox cycle parameters, sildenafil significantly reduced blood glucose and pyruvate levels and the 3-hydroxybutyrate/acetoacetate ratio, while there was no significant effect on lactate, 3-hydroxybutyrate, acetoacetate and free fatty acid levels. Symptomatic transient hypotension was observed in two women.

CONCLUSION:

The ARTERIOFIL study has shown that a single 100 mg oral dose of sildenafil had a significant effect on increase in MWT but had no significant effects on PFWT and oxygenation parameters in patients with claudication. A double-blind, prospective, randomized, multicenter study (VIRTUOSE©) is ongoing to evaluate the chronic effect of six month-long sildenafil treatment on MWT in PAD patients with claudication.

CLINICAL TRIAL REGISTRATION:

This clinical trial was registered at clinicaltrials.gov, registration. number: NCT02832570, (https://clinicaltrials.gov/ct2/show/NCT02832570).

2.6 Original aspects of the VIRTUOSE study

We hypothesise that sildenafil has a beneficial effect on walking capacity. Indeed, due to sildenafil relatively short half-life that may account for the perceived functional improvement in patients with IC. A daily oral dose of 100 mg of sildenafil is rapidly absorbed (blood peak in less than one hour) and has a half-life of about 4 to 6 hours. Indeed, one of our objectives is to improve the walking distance and benefit from the eventual analgesic effect of the molecule (as suggested by pre-clinical results) during periods of activity while also stimulating angiogenesis in ischemic areas and so avoid the need for revascularization. From this perspective, there is no need to treat during periods of inactivity such as night. This latter point is expected to result in a decrease in side effects such as orthostatic hypotension during periods of inactivity (Night time). We aim to improve walking capacity with the support of a drug devoted to functional

improvement and so modifying life-style and changing from a vicious circle of inactivity to a virtuous circle of improved activity (Figure 4).

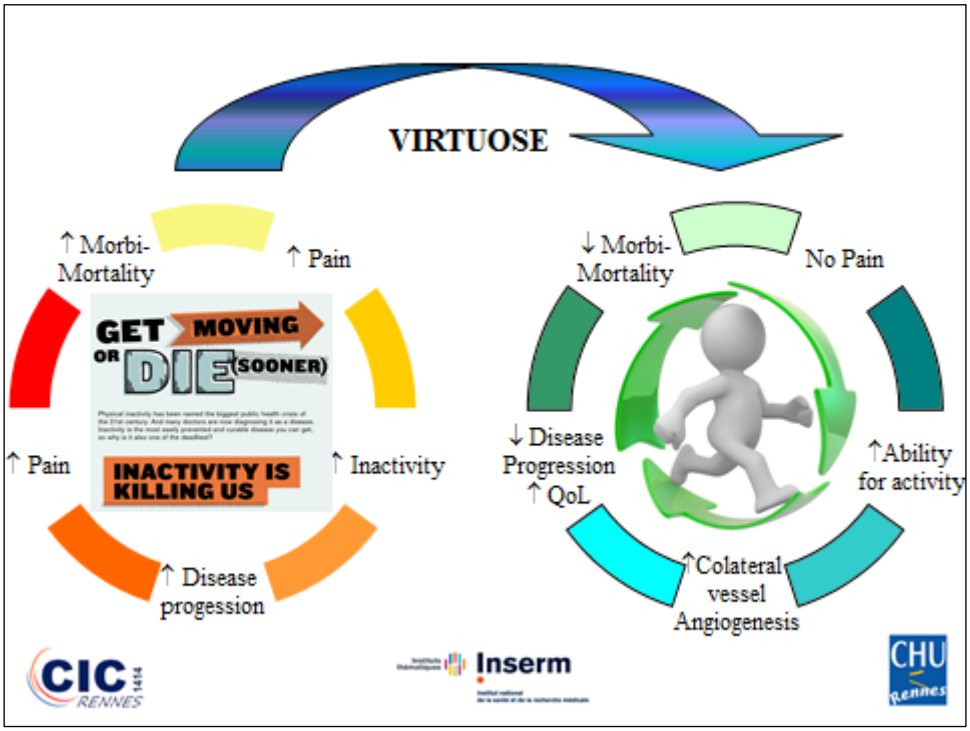


Figure 6: From a Vicious circle to a Virtuous circle with VIRTUOSE

2.7 Summary of benefits, if applicable, and of foreseeable and known risks for the person who is a subject in the research study

2.7.1 Benefits

2.7.1.1 Individual benefits

Firstly, no therapy has proven effective on claudication and improvement of walking distance in patients with claudication. The use of sildenafil for this indication is expected to represent an undeniable benefit for the patient. We expect that a significant number of patients treated with sildenafil will be sufficiently improved compared to the control group to not require surgery anymore. Further, improving quality of life of patients suffering a chronic debilitating disease is a major issue in vascular medicine. It is expected that the treatment may help patient change from a vicious circle (pain > inactivity > disease progression > pain > increased morbi-mortality) to a virtuous circle (no pain > improved ability for activity > collateral vessel development > slowing of disease progression > decreased morbi-mortality)[11].

Secondly, as for rehabilitation, we expect that functional result of patients who will nevertheless be revascularized will be higher in sildenafil treated patients than in the placebo group.

2.7.1.2 Collective benefit

The cost of treatment of symptomatic PAD is high regardless of the technique used (reconstructive surgery, thrombolysis or amputation)[98].

The increase in morbidity and mortality has high direct costs: diagnosis requires very sophisticated techniques that only trained multidisciplinary teams can use, and the use of restorative vascular surgery proves at least as onerous as that of conventional surgery. Indirect costs are likely higher, though little is still known. Thus, the cost of social and professional impact has not been assessed although the treatment can lead to unemployment that may exceed 100 days over six months of follow up.

Moreover revascularization treatments are not without complications such as surgical site infection (SSI), which is one of the most common postoperative complications after vascular reconstruction, resulting in significant morbidity and hospital readmission[98].

It is expected that the results from VIRTUOSE trial will produce:

- Decrease in direct and indirect costs for society
- Decrease in social and professional costs
- Decrease in hospital readmission following complication after surgery
- Decrease in morbi-mortality
- Decrease in cardiovascular complications
- A new paradigm in the medical treatment of IC and PAD
- Improvement in quality of life.
- Transient help to attain a virtuous circle

2.7.2 Risks

The risks associated with the study are mainly related to potential secondary and related adverse effect of sildenafil and marginally those of the diagnostic methods (TcpO2 & walking treadmill test, respiratory function tests, Arterial stiffness and compliance, central blood pressure, vascular resistance) used in the follow-up period.

Risks related to <u>sildenafil</u> have been studied extensively in the treatment of erectile dysfunction and for the treatment of pulmonary arterial hypertension. Apart from absolute restrictions and relatively well-tolerated side effects, risks related to the use of this treatment are low. Many side effects are reported according to the "product characteristics notice" (see appendix), but among frequent or severe effects are:

- Hypotension, dizziness. Hypotension is of particular risk during the night (orthostatic hypotension) with an increased risk of fallings. The use of a single morning intake as proposed in the VIRTUOSE study and should significantly reduce this risk, due to the short half-life of the molecule.
- Headache, redness of the face (skin redness), dyspepsia (heartburn) and limb pain are frequent but according to regular prescribers are rarely responsible for discontinuation of treatments.
- Stroke or infarction or cardiac arrhythmia or death. Such events are frequent in PAD patients regardless of the treatment. We expect to find a lower rate in sildenafil treated patients than in the placebo group through the shift from vicious to virtuous mechanisms allowed by the improved walking ability.
- Most visual problems are non-severe (color vision problems) although very rare but unpredictable complication have been observed (non-arteritic anterior ischemic optic neuropathy: NAION).
- Cases of prolonged erection and priapism have been reported since marketing in patients receiving sildenafil. Sildenafil should be used with caution in patients with an anatomical malformation of the penis (such as angulation, corpus cavernosum sclerosis or Peyronie's disease) or in patients with conditions that may predispose them to priapism (such as sickle cell disease, multiple myeloma, or leukemia).

  This risk may be increased by daily use of sildenafil. However, it should be noted that most patients with AOMI are also at risk for erectile dysfunction. The urology departments of the associated investigative centers will be informed of the VIRTUOSE study if a patient presents to the emergency department with priapism.

  Patients will be invited to contact the emergency department for prompt treatment.

  If an erection lasts longer than 4 hours, the patient should immediately seek medical assistance.
- Other side effects are rare or minor (headache, nasal congestion, myalgia, nausea, skin rash, palpitations, epistaxis, abdominal pain) and according to regular prescribers are rarely responsible for stopping treatment.

In general, particular care will be taken to respect the contraindications and precautions for use of the product: leukemia, sickle cell disease, multiple myeloma, severe liver damage, severe hypotension, stroke, or myocardial infarction less than 3 months old and patients under treatment of nitrates, imidazole derivatives, or ritonavir.

Women of childbearing potential and contraception in women:

Due to a lack of data in pregnant women, sildenafil is not recommended in women of childbearing potential unless appropriate contraceptive measures are used. Effective contraception (estrogen-progestin or progestin-only pill, implant, IUD, etc.) will be necessary for the duration of the study. Each patient of childbearing potential who agrees to participate in the study will be required to commit to effective contraception during the study.

Pregnancy:

There are no data from the use of sildenafil in pregnant women. Animal studies do not show direct or indirect harmful effects on pregnancy and embryo-fetal development. Studies in animals have shown postnatal developmental toxicity.

Due to a lack of data, sildenafil should not be used in pregnant women unless clearly necessary. Pregnant women cannot be included in VIRTUOSE.

Breast-feeding:

There are no specific studies in breastfeeding women. Data from a lactating woman showed excretion of sildenafil and its active metabolite, n-desmethylsildenafil, at very low concentrations in human milk. No clinical data are available on adverse reactions in breast-fed infants. Nevertheless, the amounts ingested would not a priori be likely to cause undesirable effects in the breast-fed newborn. Prescribers should consider the clinical benefit of sildenafil for the mother and the potential risk of side effects in the nursing child.

Fertility:

Conventional preclinical fertility studies have not revealed any particular risk for humans.

Contraindication:

Hypersensitivity to the active substance or to any of the excipients. Given the knowledge of its mode of action at the level of the nitrogen monoxide / cyclic guanosine monophosphate (cGMP) pathway, a potentiation of the hypotensive effects of nitrates by sildenafil has been demonstrated; its concomitant administration with nitrogen monoxide donors (such as amyl nitrite) or with nitrates, in any form whatsoever, is therefore contraindicated. Co-administration of PDE5 inhibitors, such as sildenafil, with guanylate cyclase stimulators, such as riociguat, is contraindicated due to the risk of symptomatic hypotension. Medicines used in the treatment of

erectile dysfunction, including sildenafil, should not be used in men for whom sexual activity is not recommended (e.g., patients with severe cardiovascular problems such as unstable angina or unstable angina). severe heart failure). Sildenafil is contraindicated in patients with loss of vision in one eye due to anterior non-arteritic ischemic optic neuropathy (NAION), whether or not this event was associated with previous exposure to a PDE5 inhibitor. The safety of sildenafil has not been studied in the following subgroups of patients: severe hepatic impairment, hypotension (blood pressure <90/50 mm Hg), recent history of stroke or myocardial infarction and cases of known hereditary degenerative disorders of the retina such as retinitis pigmentosa (a minority of these patients have genetic disorders of retinal phosphodiesterases).

Overdose:

In studies in volunteers receiving single doses up to 800 mg, side effects were the same as at lower doses, but their incidence and severity were increased. Doses of 200 mg do not provide greater efficacy, but the incidence of side effects (headache, flushing of the face, dizziness, dyspepsia, nasal congestion, visual disturbances) was increased. In the event of an overdose, the usual symptomatic treatment measures should be implemented as needed. Kidney dialysis is not expected to accelerate the clearance of sildenafil, as it is highly bound to plasma proteins and not excreted in the urine.

Risk related to the treadmill test and TcpO2 measurements are marginal.

According to our experience, the risk of fall during a walking treadmill test is lower than 1 out of 5,000 tests. Chest pain or arrhythmias are criteria to stop the walking test. The prevalence of coronary lesions is high in PAD patients with possible appearance during the test of a severe discomfort or chest pain that suggests the presence of coronary artery disease. If occurring, such events will be handled by the physician in charge, and the review will be communicated immediately to the patient's physician and surgeon for usual treatment. Concerning the measurement of TcpO2, to our knowledge no skin burns have been reported when 44.5 °C temperature is applied to the skin for short periods. There is no causticity of the electrolyte should the membrane rupture. Allergy to adhesives is rare and has never led to complication in over 10,000 subjects' on-going rest or exercise measurements.

For patients who will be included at Rennes University Hospital in a sub-study:

- Risk related to the Laser Speckle Contrast Imaging (LSCI) is marginal

- Risk related to pOpmetre® measurements is marginal
- Risk related to Finometer® measurements is marginal
- Risk related to pulmonary function testing and diffusion capacity of the lungs for carbon monoxide (DLCO) assessment is marginal. This examination is routinely performed in our centre without any side effects reported.
- Risk related to taking blood samples is innocuous and, in any case, temporary (bruising, venitis at the point of puncture, vagal discomfort, pain).

2.7.3 Benefit / risk ratio

We estimate that the risks and the predictable constraints for the patients participating in the study, described in paragraphs above are reasonable and acceptable with regard to the expected benefits.

2.8 Description and rationale for route of administration, dosage, administration regimen and duration of treatment

Drug administration is detailed in chapter 6.

2.9 Statement indicating that the study will be conducted in compliance with the protocol as well as with good clinical practices and legislative and regulatory conditions in force

The sponsor and the investigator also agree that this study will be conducted:

- in compliance with the protocol,
- in compliance with local and international good clinical practice currently in force,
- in compliance with legislative and regulatory conditions currently in force in participating countries.

2.10 Description of the studied population

The selection procedures are described in chapter 5.

3 STUDY OBJECTIVES

3.1 Primary objective

To determine whether sildenafil, compared with placebo, improve the walking capacity at week 24 of patients with intermittent claudication.

3.2 Secondary objectives

To determine whether sildenafil, compared with placebo, from baseline:

- prevents surgical revascularization procedure at weeks 24 and 48;
- improves the walking capacity at week 48;
- improves the time to first event;

An "EVENT" is defined as either (1) major adverse cardiovascular events (MACE; including vascular deaths, non-fatal myocardial infarction and non-fatal stroke), (2) leg amputations, (3) Non Cardiovascular death.

Event-free survival is defined as the time from inclusion to the first documented event. If no event is observed, event-free survival is defined as the delay of follow-up.

- improves the quality of life at weeks 12, 24 and 48;
- improves the exercise oximetry results at weeks 12, 24 and 48;
- improves endothelial function at weeks 12, 24 and 48;
- improves pulmonary function and diffusion capacity of the lungs for carbon monoxide (DLCO) at week 24;
- improves arterial stiffness (Pulse Wave Velocity) and Central Blood Pressure with pOpmetre® at weeks 12, 24 and 48;
- improves arterial compliance and vascular resistance with Finometer® at weeks 12, 24 and 48;
- changes metabolomic signature and biological parameters between baseline and week 24;

To evaluate:

- compliance with the treatment.
- tolerance and perceived symptoms.

4 STUDY DESIGN

4.1 Precise statement of primary evaluation criteria and secondary evaluation criteria, if applicable

4.1.1 Primary outcome measure

Absolute change of the absolute claudication distance (ACD) from baseline to week 24. ACD is measured in meters at baseline and week 24 by walking on a treadmill until the point when the subject is unable to walk anymore due to pain in the leg.

For the treadmill tests, walking will be conducted up to the maximum walking ability (not only the appearance of a pain) or to a maximum of 15 minutes.

4.1.2 Secondary outcomes measures

- Rate of patients with surgical revascularization at weeks 24 and 48;
- Absolute change of the ACD from baseline to week 48;

For the treadmill tests, walking will be conducted up to the maximum walking ability (not only the appearance of a pain) or to a maximum of 15 minutes.

- Event free survival (EFS) from baseline until week 48;

An "EVENT" is defined as either (1) major adverse cardiovascular events (MACE, including vascular deaths, non-fatal myocardial infarction and non-fatal stroke), (2) leg amputations, (3) non-cardiovascular death.

Event-free survival is defined as the time from inclusion to the first documented event. If no event is observed, event-free survival is defined as the delay of follow-up.

- SF36 questionnaire at baseline, weeks 12, 24 and 48;
- Peripheral Artery Questionnaire at baseline, weeks 12, 24 and 48;
- Change in exercise oximetry results between baseline and weeks 12, 24 and 48;
- Change in endothelial function between baseline and weeks 12, 24 and 48;
- Change in pulmonary function and diffusion capacity of the lungs for carbon monoxide (DLCO) at week 24 from baseline;
- Changes in arterial stiffness (Pulse Wave Velocity) and Central Blood Pressure with pOpmetre® between baseline and weeks 12, 24 and 48;
- Changes in arterial compliance and vascular resistance with Finometer® between baseline and weeks 12, 24 and 48;
- Change in metabolomics signature between baseline and week 24;

- Compliance with the treatment.
- Tolerance and perceived symptoms.

4.2 Description of study methodology

4.2.1 Experimental design

It is a phase III, National, Multi-centre, Prospective, Randomized, Double-blinded, Placebo-Controlled trial with two parallel group.

4.2.2 Conduct of study

4.2.2.1 Inclusion (V0)

The study will be proposed to the patients followed in the vascular department for their pathology in each of the anticipated University / Regional Hospital Centre participating in this trial. Patient's eligibility is assessed by the investigator from their medical records, according to different criteria for inclusion and non-inclusion that was evaluated during their normal medical visit.

The inclusion phase will consist of a two-step process. **A pre-inclusion visit** made of an interview, a traditional clinical examination, and an initial assessment on Treadmill test, with exercise oximetry if available in the center.

This visit will allow:

- a clinical evaluation, including evaluation of tobacco and drugs use;
- collecting pathology history and ongoing treatment apart from studied drug;
- delivering explanatory information on the trial;
- checking the patients meet the eligibility criteria, in particular:

  * Patients with peripheral artery disease (ABI ≤ 0.90 or TBI ≤ 0.70 or postexercise ABI decrease of 18.5% from rest or ABI Exercise TcPO2 with DROPmin ≤ -15 mmHg) reporting stable limiting claudication despite optimal medical treatment (Antiplatelet / Direct Oral Anticoagulant + Lipid Lowering Drugs + AT2 antagonists / ACE Inhibitors, unless contra-indication) and advice to walk for at least 4 weeks.

  * Patients with a walking capacity lower or equal to 500 meters on treadmill test. The treadmill test is performed routinely in the assessment of walking capacity of PAD patients.

- Delivering an information document along with the informed consent form.

- Performing a hepatic and renal biological assessment (TP, Total bilirubin, Albumin, Creatinine, glomerular filtration rate (GFR) with the CKD-EPI method).
- Performing a pregnancy test, for all women of childbearing age.

If the patient agreed to participate into this research **an inclusion visit** will be scheduled within five weeks. For all patients meeting the eligibility criteria, the visit will consist of:

- the signature of the informed consent (if not done at the inclusion visit);
- the randomization process.
- a clinical evaluation, including evaluation of tobacco and drugs use;
- an ECG;
- collecting clinical history;
- a Treadmill test (with oximetry if available in center);
- A 6 minuts test
- ABI and post-effort ABI measurements
- given recommendation to walk;
- collecting pathology evolution and ongoing treatment apart from studied drug;
- filling in the SF-36 questionnaire and the Peripheral Artery Questionnaire by patients evaluating the quality of life;
- dispensing treatment for 4 weeks (sildenafil or placebo of sildenafil).

If the pre-inclusion visit and the inclusion visit are both carried out on the same day or within a maximum interval of 5 days, the walking test on the treadmill and the SPI measurements will only be done once.

Only for patient who will be included at Rennes University Hospital:

- Blood Test (Standard and Metabolomics)
- Endothelial function assessment by LSCI
- Arterial Stiffness (Pulse Wave Velocity) and Central Blood Pressure with pOpmetre®
- Arterial compliance (Cwk) and vascular resistance (TPR) with Finometer®
- Pulmonary Function testing and diffusion capacity of the lungs for carbon monoxide (DLCO) assessment

4.2.2.2 Randomization

The randomization will be stratified by centre and balanced 1:1. The Biometrics Unit of the Clinical Investigation Centre Inserm 1414 in Rennes will generate the randomization list for allocation to the sildenafil and placebo groups.

4.2.2.3 Experimental group

- Sildenafil citrate 140 mg/day (single morning dose of 140 mg) in addition to usual treatment for a total duration of 24 weeks.
  Sildenafil citrate 140 mg/day is equal to citrate base 100 mg/day.

4.2.2.4 Control group

- Placebo (single morning dose) in addition to usual treatment for a total duration of 24 weeks.

Treatment will be proposed in addition to optimal treatment (Antiplatelet / Direct Oral Anticoagulant + Lipid Lowering Drugs + AT2 antagonists / ACE Inhibitors; unless contra-indicated) and advice to walk.

4.2.2.5 Monitoring of patients

Phone call at 7 and 14 days (P1 and P2)

After delivering treatment at inclusion visit for a 4-weeks treatment period, phone call will be carried out at 7 and 14 days focusing on evaluation of tobacco and drugs use, tolerance, compliance with the treatment, eventual side effects, pathology evolution and ongoing treatment apart from studied drug. Moreover, walking recommendations will be given to the patient.

Visit 1 (V1 - week 4)

First follow up visit at week 4 will consists of a clinical evaluation focusing on:

- a clinical evaluation, including evaluation of tobacco and drugs use,
- tolerance and eventual side effects,
- compliance,
- eventual side effects,
- pathology evolution and ongoing treatment apart from studied drug.

Walking recommendations will be given to the patient. If no major side effect is found during this clinical evaluation, the study drug will be delivered for an additional 8 weeks.

Phone call at 8 weeks (P3)

Phone call will be carried out at 8 weeks focusing on the same points as the phone call at 7 and 14 days.

Visit 2 (Week 12)

Patients will be evaluated at week 12 (second follow-up visit) for persistent or non-persistent indication for revascularization and considered for revascularization if needed. In parallel, this visit will consist for all patients of:

- a clinical evaluation, including evaluation of tobacco and drugs use;
- tolerance,
- compliance,
- eventual side effects,
- an ECG;
- a Treadmill test (with oximetry if available in center);
- ABI and post-effort ABI measurements
- given recommendation to walk;
- collecting pathology evolution and ongoing treatment apart from studied drug;
- filling in the SF-36 questionnaire and the Peripheral Artery Questionnaire by patients evaluating the quality of life.

Only for patient who will be included at Rennes University Hospital:

- Endothelial function assessment by LSCI at visit 2 (week 12)
- Arterial Stiffness (Pulse Wave Velocity) and Central Blood Pressure with pOpmetre®
- Arterial compliance (Cwk) and vascular resistance (TPR) with Finometer®

Attention will be given to tolerance, compliance and eventual side effects. If no major side effect is found, the study drug will be delivered for an additional 12 weeks period.

Phone calls at weeks 16 and 20 (P4 and P5)

Phone call will be carried out at weeks 16 and 20 focusing on the same points as the previous phone calls.

Visits 3 and 4 (Weeks 24 and 48)

Third and fourth follow-up visits are scheduled at weeks 24 (end of treatment) and 48 (24 weeks after the end of experimental drugs).

These two visits consist of the same visit as the visit 2 but without the dispensation of treatment. The compliance and the tolerance and eventual side effects are collecting only till the visit 3.

A 6 minuts test will be realized at visit 3.

Only for patient who will be included at Rennes University Hospital:

- Blood Test (Standard and Metabolomics) at visit 3 (week 24)
- Endothelial function assessment by LSCI at visits 3 and 4 (weeks 24 and 48)
- Pulmonary Function testing including spirometry and diffusion capacity of the lungs for carbon monoxide (DLCO) assessment at visit 3 (week 24)
- Arterial Stiffness (Pulse Wave Velocity) and Central Blood Pressure with pOpmetre® at visits 3 and 4 (weeks 24 and 48)
- Arterial compliance (Cwk) and vascular resistance (TPR) with Finometer® at visits 3 and 4 (weeks 24 and 48)

The follow-up of the patients is summarised in the following flow-chart. All follow-up visits will be performed by each co-investigator centre. All phone calls will be performed by the inclusion centre.

4.2.2.6 Post-trial care of participants

At the end of their participation, patients will be followed as usually in routine care. There is no special care required following participation in the study.

4.2.2.7 Sub-study in Rennes University Hospital

For the patients who will be included at Rennes University Hospital, a sub-study will be proposed to them including:

- a blood test (standard blood test and metabolomic study) at baseline and at week 24,
- a study of the endothelial function by Laser Speckle and a study of spirometry and diffusion capacity of the lungs for carbon monoxide (DLCO) at baseline and at weeks

12, 24 and 48,

- Pulmonary Function testing including spirometry and diffusion capacity of the lungs for carbon monoxide (DLCO) assessment at baseline and at week 24,
- Arterial Stiffness (Pulse Wave Velocity) and Central Blood Pressure with pOpmetre® at baseline and at weeks 12, 24 and 48,
- Arterial compliance (Cwk) and vascular resistance (TPR) with Finometer® at baseline, and weeks 12, 24 and 48.

Blood Test (at baseline and week 24)

To give a translational research dimension to the project, a metabolomic study will be carried out.

The clinical metabolomics is expanding. It allows, by assaying many metabolites, to search for new biomarkers with a diagnostic, prognostic or therapeutic value of follow-up. It also allows a broader biological vision of pathologies.

The metabolomic approach requires a homogeneous patient population that will be compared to a control population. Multivariate statistical analyzes, once the metabolites are assayed, reveal the metabolites that distinguish the diseased population from the control population. The metabolites on or underrepresented in the biological samples of the diseased population thus become biomarkers for the pathological process. This metabolic signature, a control of the physiopathological process (example of ischemia of the lower limbs), is complex to analyze. It includes controls of pathogenic mechanisms (increase of lactate during cellular hypoxia in our example), witnesses of cellular suffering (ex: lysis of muscle cells), controls of regulatory mechanisms attempting to overcome dysfunctions induced by pathology (eg increase in anaerobic glycolysis), controls for disruption of signaling pathways (eg serotonin, adrenaline).

The simultaneous determination of many metabolites is very delicate as it is subject to many pre-analytical and analytical variations that could distort statistical analysis.

The pre-analytical aspects will therefore be closely controlled: blood samples (55 ml) under the same conditions (Clinical Investigation Unit) for all patients. The blood tubes will be sent directly to the Biological Research Center of the University Hospital of Rennes. They will be centrifuged at 2500 g for 10 minutes without brake and at + 4 ° C. Aliquots of the supernatant (plasma) will be prepared and stored at -80 ° C before the assay.

Centrifugation of tubes, aliquoting and plasma storage will be performed at the Biological Resource Center of the Rennes Hospital at -80 °C.

Once all the plasmas have been collected, the Biochemistry laboratory of Angers University

Hospital (Pr Pascal Reynier) will carry out the metabolomic analysis and the multivariate statistical analysis. The metabolic signature associated with ischemia will then be the subject of bibliographic work and physiopathological understanding before being valorised in the form of a patent or a publication. Note that the laboratory has access to different technologies allowing the confirmation of the results by another approach (chromatography of amino acids and their derivatives, analysis of organic acids, high resolution mass spectrometer Q Exactive Orbitrap Thermo Fisher).

List of metabolites that will be measured:

Absolute/DQ® p180 List of Metabolites
AB SCIEX®

Acylcarnitines (40)			
C0	Carnitine	C10:1	Decenoylcarnitine
C2	Acetylcarnitine	C10:2	Decadienylcarnitine
C3	Propionylcarnitine	C12	Dodecanoylcarnitine
C3:1	Propenoylcarnitine	C12:1	Dodecenoylcarnitine
C3-OH	Hydroxypropionylcarnitine	C12-DC	Dodecanedioylcarnitine
C4	Butyrylcarnitine	C14	Tetradecanoylcarnitine
C4:1	Butenylcarnitine	C14:1	Tetradecenoylcarnitine
C4-OH (C3-DC)	Hydroxybutyrylcarnitine	C14:1-OH	Hydroxytetradecenoylcarnitine
C5	Valerylcarnitine	C14:2	Tetradecadienylcarnitine
C5:1	Tiglylcarnitine	C14:2-OH	Hydroxytetradecadienylcarnitine
C5:1-DC	Glutaconylcarnitine	C16	Hexadecanoylcarnitine
C5-DC (C6-OH)	Glutarylcarnitine* (Hydroxyhexanoylcarnitine)	C16:1	Hexadecenoylcarnitine
C5-M-DC	Methylglutarylcarnitine	C16:1-OH	Hydroxyhexadecenoylcarnitine
C5-OH (C3-DC-M)	Hydroxyvalerylcarnitine (Methylmalonylcarnitine)	C16:2	Hexadecadienylcarnitine
C6 (C4:1-DC)	Hexanoylcarnitine (Fumarylcarnitine)	C16:2-OH	Hydroxyhexadecadienylcarnitine
C6:1	Hexenoylcarnitine	C16-OH	Hydroxyhexadecanoylcarnitine
C7-DC	Pimelylcarnitine	C18	Octadecanoylcarnitine
C8	Octanoylcarnitine	C18:1	Octadecenoylcarnitine
C9	Nonaylcarnitine	C18:1-OH	Hydroxyoctadecenoylcarnitine
C10	Decanoylcarnitine	C18:2	Octadecadienylcarnitine

Amino Acids and Biogenic Amines (40)			
Ala	Alanine	Ac-Orn	Acetylornithine
Arg	Arginine	ADMA	Asymmetric dimethylarginine
Asn	Asparagine	SDMA	Symmetric dimethylarginine
Asp	Aspartate	total DMA	Total dimethylarginine
Cit	Citrulline	alpha-AAA	alpha-Aminoadipic acid
Gln	Glutamine	Carnosine	Carnosine
Glu	Glutamate	Creatinine	Creatinine
Gly	Glycine	Histamine	Histamine
His	Histidine	Kynurenine	Kynurenine
Ile	Isoleucine	Met-SO	Methioninesulfoxide
Leu	Leucine	Nitro-Tyr	Nitrotyrosine
Lys	Lysine	OH-Pro	Hydroxyproline
Met	Methionine	PEA	Phenylethylamine
Orn	Ornithine	Putrescine	Putrescine
Phe	Phenylalanine	Sarcosine	Sarcosine
Pro	Proline	Serotonin	Serotonin
Ser	Serine	Spermidine	Spermidine
Thr	Threonine	Spermine	Spermine
Trp	Tryptophan	Taurine	Taurine
Tyr	Tyrosine		
Val	Valine		

Validated for human blood plasma — modified 2012-04-26

Hexoses (1)			
H1	Hexose		

Sphingolipids (15)			
SM (OH) C14:1	SM C18:0	SM (OH) C22:1	SM (OH) C24:1
SM C16:0	SM C18:1	SM (OH) C22:2	SM C26:0
SM C16:1	SM C20:2	SM C24:0	SM C26:1
SM (OH) C16:1	SM C22:3	SM C24:1	

Glycerophospholipids (90)			
lysoPC a C14:0	PC aa C34:1	PC aa C42:0	PC ae C38:2
lysoPC a C16:0	PC aa C34:2	PC aa C42:1	PC ae C38:3
lysoPC a C16:1	PC aa C34:3	PC aa C42:2	PC ae C38:4
lysoPC a C17:0	PC aa C34:4	PC aa C42:4	PC ae C38:5
lysoPC a C18:0	PC aa C36:0	PC aa C42:5	PC ae C38:6
lysoPC a C18:1	PC aa C36:1	PC aa C42:6	PC ae C40:1
lysoPC a C18:2	PC aa C36:2	PC ae C30:0	PC ae C40:2
lysoPC a C20:3	PC aa C36:3	PC ae C30:1	PC ae C40:3
lysoPC a C20:4	PC aa C36:4	PC ae C30:2	PC aa C40:4
lysoPC a C24:0	PC aa C36:5	PC ae C32:1	PC ae C40:5
lysoPC a C26:0	PC aa C36:6	PC ae C32:2	PC ae C40:6
lysoPC a C26:1	PC aa C38:0	PC ae C34:0	PC ae C42:0
lysoPC a C28:0	PC aa C38:1	PC ae C34:1	PC ae C42:1
lysoPC a C28:1	PC aa C38:3	PC ae C34:2	PC ae C42:2
PC aa C24:0	PC aa C38:4	PC ae C34:3	PC ae C42:3
PC aa C26:0	PC aa C38:5	PC ae C36:0	PC ae C42:4
PC aa C28:1	PC aa C38:6	PC ae C36:1	PC ae C42:5
PC aa C30:0	PC aa C40:1	PC ae C36:2	PC ae C44:3
PC aa C30:2	PC aa C40:2	PC ae C36:3	PC ae C44:4
PC aa C32:0	PC aa C40:3	PC ae C36:4	PC ae C44:5
PC aa C32:1	PC aa C40:4	PC ae C36:5	PC ae C44:6
PC aa C32:2	PC aa C40:5	PC ae C38:0	
PC aa C32:3	PC aa C40:6	PC ae C38:1	

Regarding the analytical aspects, we chose to use a targeted metabolomic approach using a standardized kit developed by the Austrian company Biocrates Life Sciences. This kit (AbsoluteIDQ®p180) benefits from internal standards, quality controls, a dedicated analysis software and numerous publications validating its use in clinical biology. This kit has been the subject of a method validation in the laboratory that has all the equipment and skills necessary for its use. The analysis, carried out on a QTRAP LC / MS / MS 5500 mass spectrometer (AB Sciex), makes it possible to assay 188 metabolites (acyl-carnitines, amino acids, biogenic amines, glycerophospholipids, sphingolipids and sugars) starting from 20 microliters of plasma and more a very wide range of concentrations. His interest is to simultaneously measure polar molecules and lipids (lipidomics).

Patients included in Rennes will be collected for the study of metabolomics.

Standard examinations such as blood count, fibrinogen will be done in hematology laboratory; ultra-sensitive C-reactive protein, Urea, Creatinine, total cholesterol, LDL-cholesterol, HDL-

cholesterol, triglycerides, blood-glucose, glycated hemoglobin (HbA1c), insulinemia, Vascular Endothelial Growth Factor, Lactate, Pyruvate, Ammoniac, Creatine Phosphokinase, 3 hydoxybutyrate, acetoacetate, free fatty acid, sodium, potassium, chlorine, bicarbonates, albumin, uric acid, ASAT, ALAT, GGT, PAL, ApoA1, ApoB, Vitamin B9, B12, B6, Total proteins, LpPLA2, Homocysteine, oxidized LDL, Lp (a) and Profile of acyles-carnitine will be done in biochemistry laboratory and also because the metabolites are not integrated in the p180 kit. The increase in lactate will bring a positive control of ischemia.

Endothelial function by Laser Speckle (at baseline, weeks 12, 24 and 48):

Cutaneous microcirculation is considered as a mirror of generalized vascular function and is a useful translational model for investigating mechanisms of cutaneous physiology and cutaneous pathophysiology due either to skin disease or to other diseases. Various microvascular tests for some of these, though not validated, have been used to assess both neurovascular abnormalities and endothelial dysfunction[99]. Several dedicated tests are available and have been well-described for assessing the various mechanisms of vasodilation such as the NO-mediated pathway. The aims of the present sub-study will be to use two dedicated microvascular tests to assess neurovascular functions and endothelial functions in PAD patients with claudication under placebo and compared with sildenafil at baseline and at weeks 12, 24 and 48. The duration of each visit will be 1 hour.

PORH (endothelial-dependent and endothelial-independent pathway):

The PORH for exploring neurovascular interactions and probably the endothelium-derived hyperpolarizing factor (EDHF) was measured by producing ischemia for 3 minutes with a blood pressure cuff inflated to 50 mm Hg over the subject's systolic blood pressure[100].

Local thermal hyperemia (neurovascular and endothelial-dependent pathway):

The hyperemic response to heat was evaluated by using a skin-heating probe filled with deionized water and heated to 44 °C to assess heat-induced vasodilation. Heat-induced vasodilation is due to nerve fibers for the early peak and to the NO-dependent pathway and the EDHF-dependent pathway for the late plateau[101].

Pulmonary Functional Tests (PFTs) - Spirometry and diffusion capacity of the lungs for carbon monoxide (DLCO) (at baseline and week 24):

PFT included spirometry and single-breath carbon monoxide diffusing capacity (DLCO) all measured by MasterScreen Body apparatus (Jaeger, Wurzburg, Germany). Spirometry measurements were forced vital capacity (FVC) and forced expiratory volume during the first second of expiration (FEV1). Best values from all spirometry trials were selected and were expressed as both percent of predicted normal (% pred) and as absolute values.

Arterial Stiffness (Pulse Wave Velocity) and Central Blood Pressure with pOpmetre® (at baseline and at weeks 12, 24 and 48)
See below.

Arterial compliance (Cwk) and vascular resistance (TPR) with Finometer® (at baseline, and weeks 12, 24 and 48)
See below.

Flow-chart

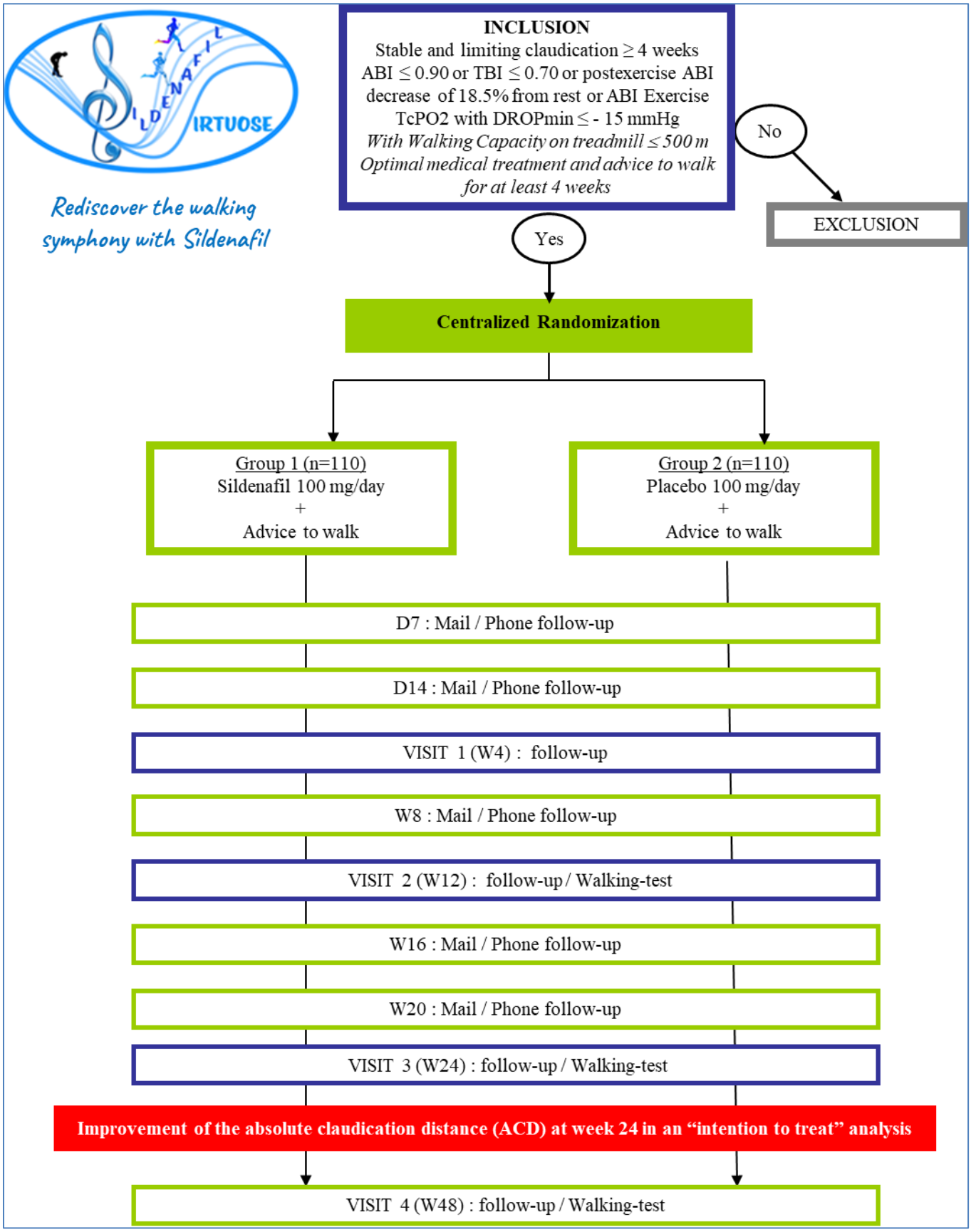
If all the exams of a visit cannot be carried out on the same day (in particular for the Rennes center who realizes the sub-study), they can be carried out over two half-days with a maximum interval of 5 days. If so, for the inclusion visit, randomization and treatment dispensing will be done after all the examinations have been carried out, i.e. at the end of the 2nd half-day.

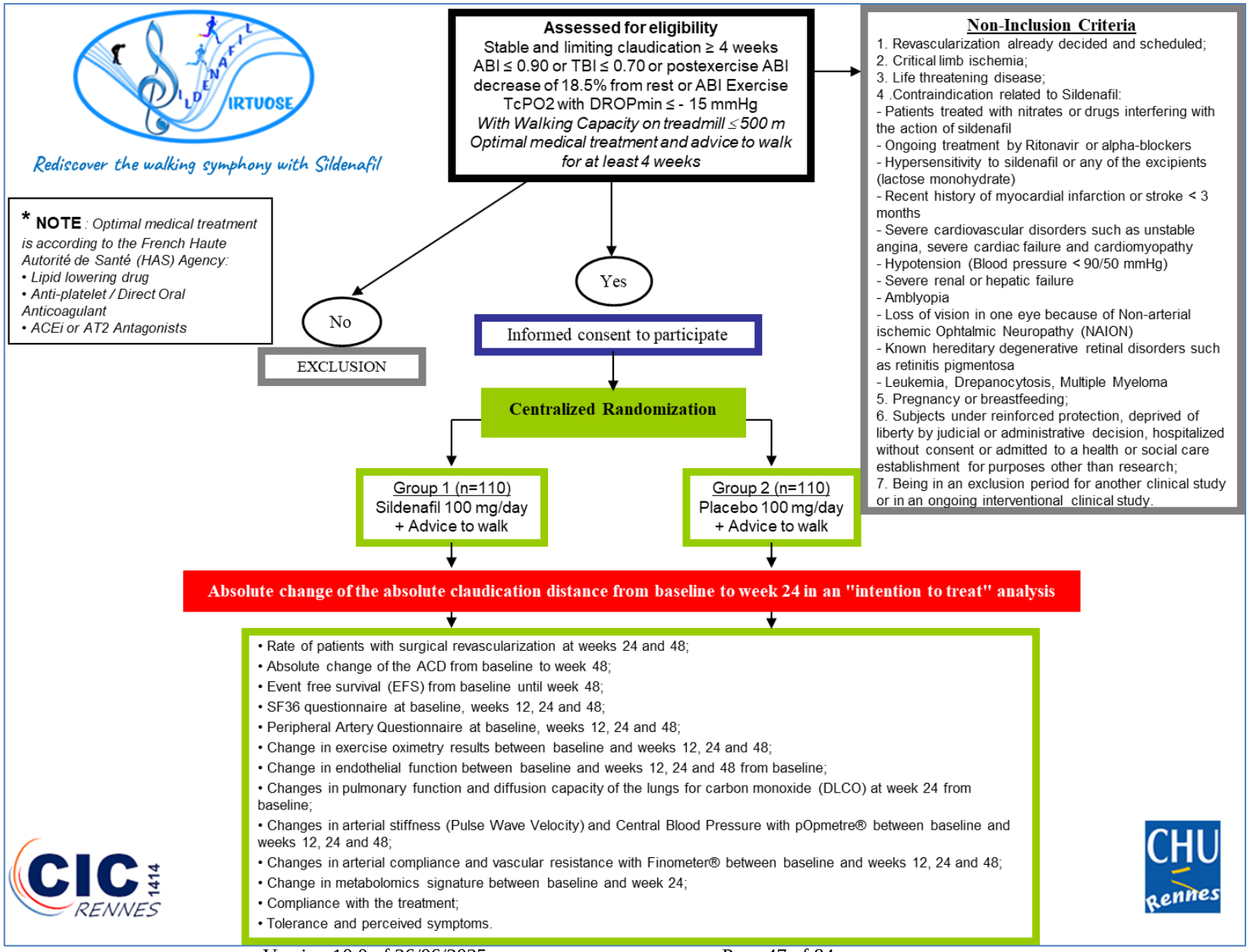
Visit (V) *or Mail/Phone call (P)*	*Pre-inclusion visit*		*Inclusion Visit V0*		*P1*		*P2*		**V1**		*P3*		**V2**		*P4*		*P5*		**V3**		**V4**	
	Within 5 weeks before V0		**Baseline**		*7 days +/- 3d*		*14 days +/- 5d*		**4 weeks +/- 4d**		8 weeks +/-7 d		**12 weeks +/-8d**		**16 weeks +/- 8d**		**20 weeks +/- 8d**		**24 weeks +/- 12d**		**48 weeks +/- 15d**	
	ARC	**AAR**	**ARC**	**AAR**	**ARC**	**AAR**	**ARC**	**AAR**	**ARC**	**AAR**	**ARC**	**AAR**	**ARC**	**AAR**	**ARC**	**AAR**	**ARC**	**AAR**	**ARC**	**AAR**	**ARC**	**AAR**
Eligibility criteria, including pregnancy test if applicable		X		X																		
Information		X																				
Signature of the informed consent[1]		X		X																		
Hepatic and renal biological assessment	X																					
Pregnancy test (if applicable)		X																				
Inclusion, randomization				X																		
Clinical history				X																		
Clinical evaluation including evaluation of tobacco and drugs use	X			X		X[2]		X[2]		X		X[2]		X		X[2]		X[2]		X		X
ABI and post-effort ABI measurements	X			X										X						X		X
ECG				X										X						X		X
Treadmill test (with oximetry if available in center)	X			X										X						X		X
6 minuts test				X																X		
Recommendation to walk				X		X		X		X		X		X		X		X		X		X
Dispensing treatment				X						X				X								
Pathology history/evolution and ongoing treatment apart from studied drug		X		X		X		X		X		X		X		X		X		X		X
Questionnaires on events occurrence						X		X		X		X		X		X		X		X		X
Compliance with treatment accounting						X		X		X		X		X		X		X		X		
Tolerance, adverse events						X		X		X		X		X		X		X		X		
Peripheral Artery Questionnaire				X										X						X		X
QoL: SF36 questionnaire				X										X						X		X
Endothelial function (LSCI)[3]				X										X						X		X
Spirometry and diffusion capacity of the lungs for carbon monoxide (DLCO)[3]				X																X		
Popmètre et Finometre[3]				X										X						X		X
Metabolomic[3] including:																						
Standard examinations[3]			X																X			
Blood Sampling[3]				X																X		

ARC: Act of Routine Care **AAR**: Act Added by the Research

[1] Informed consent signed at pre-inclusion or inclusion visit [2] Only evaluation of tobacco and drugs use during phone calls [3] Only performed in Rennes

Clinical trial scheme




Rediscover the walking symphony with Sildenafil
Assessed for eligibility
Stable and limiting claudication ≥ 4 weeks
ABI ≤ 0.90 or TBI ≤ 0.70 or postexercise ABI decrease of 18.5% from rest or ABI Exercise TcPO2 with DROPmin ≤ - 15 mmHg
With Walking Capacity on treadmill ≤ 500 m
Optimal medical treatment and advice to walk for at least 4 weeks
Non-Inclusion Criteria
1. Revascularization already decided and scheduled;
2. Critical limb ischemia;
3. Life threatening disease;
4 .Contraindication related to Sildenafil:
- Patients treated with nitrates or drugs interfering with the action of sildenafil
- Ongoing treatment by Ritonavir or alpha-blockers
- Hypersensitivity to sildenafil or any of the excipients (lactose monohydrate)
- Recent history of myocardial infarction or stroke < 3 months
- Severe cardiovascular disorders such as unstable angina, severe cardiac failure and cardiomyopathy
- Hypotension (Blood pressure < 90/50 mmHg)
- Severe renal or hepatic failure
- Amblyopia
- Loss of vision in one eye because of Non-arterial ischemic Ophtalmic Neuropathy (NAION)
- Known hereditary degenerative retinal disorders such as retinitis pigmentosa
- Leukemia, Drepanocytosis, Multiple Myeloma
5. Pregnancy or breastfeeding;
6. Subjects under reinforced protection, deprived of liberty by judicial or administrative decision, hospitalized without consent or admitted to a health or social care establishment for purposes other than research;
7. Being in an exclusion period for another clinical study or in an ongoing interventional clinical study.
* NOTE : Optimal medical treatment is according to the French Haute Autorité de Santé (HAS) Agency:
• Lipid lowering drug
• Anti-platelet / Direct Oral Anticoagulant
• ACEi or AT2 Antagonists
No
EXCLUSION
Yes
Informed consent to participate
Centralized Randomization
Group 1 (n=110)
Sildenafil 100 mg/day
+ Advice to walk
Group 2 (n=110)
Placebo 100 mg/day
+ Advice to walk
Absolute change of the absolute claudication distance from baseline to week 24 in an "intention to treat" analysis
• Rate of patients with surgical revascularization at weeks 24 and 48;
• Absolute change of the ACD from baseline to week 48;
• Event free survival (EFS) from baseline until week 48;
• SF36 questionnaire at baseline, weeks 12, 24 and 48;
• Peripheral Artery Questionnaire at baseline, weeks 12, 24 and 48;
• Change in exercise oximetry results between baseline and weeks 12, 24 and 48;
• Change in endothelial function between baseline and weeks 12, 24 and 48 from baseline;
• Changes in pulmonary function and diffusion capacity of the lungs for carbon monoxide (DLCO) at week 24 from baseline;
• Changes in arterial stiffness (Pulse Wave Velocity) and Central Blood Pressure with pOpmetre® between baseline and weeks 12, 24 and 48;
• Changes in arterial compliance and vascular resistance with Finometer® between baseline and weeks 12, 24 and 48;
• Change in metabolomics signature between baseline and week 24;
• Compliance with the treatment;
• Tolerance and perceived symptoms.
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4.2.2.8 Follow-up parameters

Clinical history

These following data concerning the clinical history will be collected at V0 in the case report form:

- Addiction to smoking and drugs
- Diabetes
- Hypercholesterolemia
- Arterial hypertension
- Coronary heart disease
- Cerebrovascular events
- Lower limb revascularization
- Obstructive Sleep Apnoea Syndrome
- Chronic respiratory diseases
- Cancer or autoimmune disease
- Renal and hepatic insufficiency
- Myocardial infarction
- Other(s) cardiovascular disease
- Other(s) significant events.

Clinical evaluation

Clinical evaluation will be performed at visit V0, V1, V2, V3 and V4.

- Ankle to Brachial Index (ABI) or Toe to Brachial Index (TBI) and post-effort ABI (except from V1) according to the standard procedure: ABI is measured and calculated for all patient visits as the ratio of the highest systolic blood pressure in each ankle from the right and left posterior tibial or dorsalis pedis arteries, divided by the highest brachial systolic blood pressure. TBI is measured and calculated for all patient visit if ABI is not contributed between the systolic blood-pressure measured at the hallux and the brachial systolic blood-pressure (measured in both arms, retaining the higher value).
- Body Mass Index
- Cardiovascular examination
- Pulse palpation.
- Initials of first and last name (only at V0)

- Date of birth (only at V0)
- Gender (only at V0)
- Stature (only at V0)

Treadmill test

Treadmill tests will be performed during pre-inclusion visit, V0, V2, V3, V4 according to a standard technique and standard recording method. For the treadmill tests, the speed will be started at 1 km/h and gradually increased to a maximum of 3.2 km/h within 25 seconds and with a constant slope of 10%. Maximum walking time (MWT) will be calculated from the time when the speed (3.2 km/h) and slope (10%) are reached. Time for the occurrence of pain and MWT will be recorded. For the follow-up tests, walking will be conducted up to the maximum walking ability (not only the appearance of a pain) or to a maximum of 15 minutes (~ 750 meters). At minute 15, in the absence of, or in case of non-limiting, lower limb pain or discomfort, the speed and grade will be progressively increased according to previously reported procedure [(89)]. This combined constant + incremental procedure allows for the quantification of any improvement even in case of spectacular improvement leading to almost normal walking capacity. The treadmill tests will be performed in accordance with good clinical practice rules and in optimum safety conditions. Patients will be encouraged to report and rate lower limb pain and/or non-lower limb symptoms throughout and at the end of the exercise.

Exercise oximetry

Transcutaneous oximetry (TcpO2) measurements will be performed only in the investigative centers with the equipment to perform it.

Transcutaneous oximetry (TcpO2) measurements will be carried out during the treadmill tests, using Radiometer (Copenhagen, Denmark) or Perimed (Stockholm, Sweden) connected to a computer preinstalled with the homemade Oxymonitor software developed at Rennes University Hospital by Professor Yves Gandon[102]. This program automatically calculates the minimum DROP for each limb probe using the chest as a reference (the methodology has already been published).

The DROP index (Decrease from Rest of Oxygen Pressure) corresponds to the limb changes from rest, minus chest changes from rest. The main advantage of this calculation is to overcome the main limitation of the TcpO2 technique related to unpredictable transcutaneous gradient affecting the absolute starting value. DROP value is, by definition, smaller than or equal to zero. The normal range of this drop index when validated against the gold standard arteriography is

between 0 and -15 mmHg proximally as distally. The advantage of the technique is that it allows bilateral recordings, is not affected by arterial stiffness and is non-invasive. Finally, the measurements are not sensitive to cardiac arrhythmia problems, and thus overcome some of the limitations of ultrasound techniques and in particular the measurement of ABI post exercise. TcpO2, through the DROP index, allows easy and effective screening of ischemia in patients with claudication, which, to date, was not feasible.

On follow up visits, the probes will be positioned at the same place as for the inclusion test. At least three probes will be used simultaneously: one on the chest, one on the most painful areas and one contra-lateral at the same level (calves and/or buttocks). Probe temperature at the measurement site will be 44.5°C. A pre-heating period of at least 15 minutes will be used to allow stabilisation of values. Then measurement of TcpO2 will be recorded for 2 minutes at rest, throughout exercise and for a minimum of 10 minutes during recovery.

For each test, the investigator will measure:

- Relative walking distance/time: Time at which pain appears
- Maximum walking time: Time for the walking period
- Recovery time.
- Minimal value of the DROP index on each limb.

Pulse wave velocity measurements (PWV)

PWV will be performed during pre-inclusion visit, V0, V2, V3, V4 according to a standard technique and standard recording method using the pOpmetre®.

PWV can be measured on different arterial regions by measuring the propagation time of the pressure pulse from one artery to another[103]. The pOpmetre® system is based on the hypothesis that the mechanical properties of the peripheral arteries do not affect the measurement of aortic stiffness. It was developed on similar assumptions as those for brachial–ankle devices in which they need to use four cuffs to conduct the measurements. The pOpmetre® uses two photodiode sensors, similar to pulse oximeters. The sensors are positioned on the finger and the toe (same side) so that the pulpar arteries are in the scope of the infrared ray. The pOmetre® measures the transit time between the foot of the pulse waves of the finger and of the toe, presuming to approximate the aortic pulse transit time. It reflects the central aortic stiffness. pOpmetre® 300 is a class IIa, non-invasive medical device for measuring the PWV[103]. The following parameters will be recorded: Transit Time (TT), PWV, Central Systolic Blood Pressure (CSBP), Central Diastolic Blood Pressure (CDBP) and Central Mean Blood Pressure (CMBP).

Cardiovascular measurements

Cardiovascular measurements will be performed during pre-inclusion visit, V0, V2, V3, V4 according to a standard technique and standard recording method using the Finometer®.

The validated Finometer® device (Finapres Medical Systems, Amsterdam, The Netherlands) will be connected to the index finger, and after a 10-minute resting period, a 5-minute continuous measurement of resting cardiovascular parameters, including systolic blood pressure (SBP), diastolic blood pressure (DBP), mean arterial pressure (MAP), total peripheral resistance (TPR), arterial compliance (Cwk) and pulse pressure (PP). During the recording, after 2 minutes, a return-to-flow systolic calibration will be performed to provide an individual subject-level adjustment of the finger arterial pressure with the brachial arterial pressure. The highest precision in cardiovascular measurements will be obtained only after this calibration. The average of the recordings of the last minute will be used[104]. The following parameters will be recorded: SBP, DBP, MAP, TPR, Cwk and PP.

Recommendation to walk

At each visit and phone call, patients will be encouraged to perform an outdoor community-based exercise program of at least 30 minutes per day and asked about their estimated daily walking distance/time in the last week. They will be told that, if pain occurs during walking, it is better to slow down until the pain subsides rather than stop and thus continue their programs without stopping. They will also be reminded that a standardised walking route is an easy manner to record the time of onset of pain and will allow precise monitoring of their progression. Following their walking program, patients are encouraged to make the most of daily tasks in order to stay active. Lastly, as a direct link to the protocol, patients will be reminded to wait for approximately one hour after taking sildenafil or placebo, before starting their outdoor community-based exercise program.

On-going treatments apart from studied drug

On-going treatment will be retrieved from the patient's history and the patient's file.

Questionnaires on event occurrence

Event occurrence will be retrieved from the patient's history or the patient's file. If unavailable, contact will be made with the patient’s general practitioner.

Questions will specifically focus on:

- Lower limb arterial revascularization and date of the event
- Non-lower limb arterial revascularization (myocardial, carotid, etc) and date of the event
- Hospitalisation for cardiovascular events or death (cardio-vascular or non-cardiovascular), and date of the event
- Stroke or myocardial infarction or acute limb ischemia and date of the event
- Others events (Haemorrhages…) and date of the event.

Compliance with the treatment

At each visit, patients are asked to provide the blisters of the studied drugs. Similarly, during the phone call, patients will be asked to state the number of capsules remaining.

The number of capsules remaining at the date of exam is measured.

Compliance will be measured by prospective capsule count.

Initials and patient's number as well as date of dispensation are completed on the label of the bottle by the pharmacy at the time of the dispensation of treatments to the patient.

After use, empty, full or partially used blisters are brought back to the centre by the patient. The return date is completed on the label of the bottle by the pharmacy at the time of the return of treatments by the patient.

At the end of the treatment, blisters are brought back to the pharmacy in every centre.

Tolerance, adverse events

Patients are advised about potential adverse effects and requested to report any adverse event. Safety will be assessed by adverse-event monitoring, analysis, measurement of vital signs, physical examination (at 4, 12 and 24 weeks following initiation of treatment), and phone calls (at 7 days, 14 days, 8, 16 and 20 weeks following initiation of treatment).

These specific adverse events will be checked at each follow up visit and mail/phone call:

- Headache
- Dizziness
- Drowsiness
- Warmth or redness in the face, neck, or chest
- Visual problems (color vision problems, visual disturbance, blurry vision…)
- Blindness: (3/100,000 treated patients per year), monocular vision on both eyes;

- Gastrointestinal disturbances : nausea, dyspepsia, abdominal pain, dry mouth, vomiting, gastrooesophageal reflux,)
- Nasal congestion
- Epistaxis
- Rhinitis
- Hypoaesthesia
- Muscular pain
- Thoracic pain
- Palpitations
- Arrhythmia
- Stroke
- Cardiac infarct
- Ringing in the ears,
- Loss of hearing,
- Skin rash
- Hematuria
- Priapism
- Penile haemorrhage

Peripheral Artery Questionnaire

The Peripheral Artery Questionnaire (PAQ) will be performed at visit V0, V2, V3 and V4. PAQ is a 20-item questionnaire developed to meet this need by quantifying patients' physical limitations, symptoms, social function, treatment satisfaction, and quality of life. The PAQ is a valid, reliable, and responsive disease-specific measure for patients with PAD. It may prove to be useful and point in clinical trials and a potential aid in disease management.

Health-Related Quality of life: SF36 questionnaire

The short-form 36 (SF-36) will be performed at visit V0, V2, V3 and V4. This questionnaire provides a comprehensive measure of the disability degree experienced by the patient as a result of PAD[91]. All patients will independently complete the SF-36 forms. The investigator will screen for missing (no answer to an item), duplicate (two or more answers to the same item) or paradoxical (e.g. higher difficulty for an easier task) answers. Each detected error will be discussed with the patient and corrected appropriately.

Spirometry and diffusion capacity of the lungs for carbon monoxide (DLCO)

Spirometry and diffusion capacity of the lungs for carbon monoxide will be performed at visit V0 and V3.

4.3 Description of the measures taken to reduce and prevent bias

4.3.1 Randomization

The randomization will be stratified by centre and balanced 1:1. The Biometrics Unit of the Clinical Investigation Centre Inserm 1414 of Rennes will create the randomization list for allocation to the treatment groups. This list will be transmitted to the PPRIGO hospital pharmacist's consortium (Production Pharmaceutique pour la Recherche Institutionnelle du Grand Ouest) for the preparation of the study treatments.

4.3.2 Methods of blinding

The preparation of the ‘blinded’ treatments will be undertaken by the PPRIGO hospital pharmacist's consortium (Production Pharmaceutique pour la Recherche Institutionnelle du Grand Ouest) under recommended standardised conditions. PPRIGO will provide numbered and labelled blisters each containing 32 capsules of the study drug (sildenafil or placebo) according to the randomization order). All blisters will be identically labelled, with the study number being the only differentiating feature between different drug packs.

The un-blinding will be centralised with ‘Ennov clinical’ software in agreement with the principal investigator.

4.4 Description of dosage and methods of administration of the investigational medicinal products. Description of dose unit form, packaging and labelling of the investigational medicinal product(s)

The description of the medicinal product (Dosage and methods of administration, Unit form, packaging and labelling) is detailed in chapter 6.

4.5 Expected duration of participation of persons and description of chronology and of duration of all study periods including monitoring, if applicable

Anticipated Duration of Recruitment: 36 months

Duration of participation of each patient: 12 months

Duration of analysis of data: 6 months

Estimated total duration of study: 54 months

Starting with inclusion of the first patient, the sponsor has to inform without delay the local health agency and the ethics committee of the actual date of start-up of the study. The actual date of start-up = date of signature of consent form by the first person who is a subject in the study.
The date of end of the study will be transmitted by the sponsor to the ethics committee and to the local health agency within 90 days or earlier according to local regulations**.** The date of end of the study corresponds to the last visit of the last person who is a subject in the study.

4.6 Description of rules for permanent or temporary discontinuation

4.6.1 Discontinuation of participation of a person in study

Subjects can withdraw their consent and ask to withdraw from the study at any time and for whatever reason. If a subject discontinues the study before its completion, the investigator must document the reasons as completely as possible.
In fact, the investigator can temporarily or permanently discontinue the study treatment for any reason which is in the best interest of the subject.
If a subject is lost to follow-up, the investigator will make every effort to resume contact with that person.

4.6.2 Discontinuation of part or of the entire study

Unexpected events or new information pertaining to the product, in light of which the study objectives or clinical programme probably would not be achieved, can lead the sponsor to terminate the study.
In case of early permanent discontinuation of the study, the information will be sent by the sponsor within 15 days to the relevant country health agency and to the ethics committee or earlier according to local regulations.

5 SCREENING AND NON-INCLUSION OF PERSONS FROM THE STUDY

5.1 Inclusion criteria for persons who are subjects in the study

1. Patient ≥ 18 years old;

2. with peripheral artery disease ABI ≤ 0.90 or TBI ≤ 0.70 or postexercise ABI decrease of 18.5% from rest or ABI Exercise TcPO2 with DROPmin ≤ -15 mmHg reporting stable limiting claudication despite optimal medical treatment (Antiplatelet / Direct Oral Anticoagulant + Lipid Lowering Drugs + AT2 antagonists / ACE Inhibitors; unless contra-indication) and advise to walk for at least 4 weeks;
3. with a walking capacity lower or equal to 500 meters on treadmill;
4. Affiliate to a social security agency
5. Patient who has understood the protocol and signed the consent form to participate.

5.2 Non-inclusion criteria for persons who are subjects in the study

1. Revascularization already decided and scheduled;
2. Critical limb ischemia;
3. Life threatening disease;
4. Contraindication related to sildenafil:
   - Patients treated with nitrates or drugs interfering with the action of sildenafil
   - Ongoing treatment by Ritonavir or alpha-blockers
   - Hypersensitivity to sildenafil or any of the excipients (lactose monohydrate)
   - Recent history of myocardial infarction or stroke < 3 months
   - Severe cardiovascular disorders such as unstable angina, severe cardiac failure and cardiomyopathy
   - Hypotension (Blood pressure < 90/50 mm Hg)
   - Severe renal or hepatic failure
   - Amblyopia
   - Loss of vision in one eye because of Non-arterial ischemic Ophtalmic Neuropathy (NAION)
   - Known hereditary degenerative retinal disorders such as retinitis pigmentosa
   - Leukemia, Drepanocytosis, Multiple Myeloma
5. Pregnancy or breastfeeding;
6. Subjects under reinforced protection, deprived of liberty by judicial or administrative decision, hospitalized without consent or admitted to a health or social care establishment for purposes other than research;
7. Being in an exclusion period for another clinical study or in an ongoing interventional clinical study.

Any one of the above criteria will prevent the patient's inclusion in the trial.

5.3 Methods of recruitment

The study will be proposed to the patients followed in the vascular unit/department for their pathology in each of the anticipated University / Regional Hospital Centres participating in this trial. Patients fulfilling the inclusion criteria will be invited to participate in this study. The investigator will present the study to them and will give them the information documents. They will then allow the subject the necessary time to take their decision to participate.

5.4 Procedure for early permanent discontinuation of treatment corresponding to discontinuation of treatment with the investigational medicinal product, and procedure for exclusion from the study corresponding to discontinuation of treatment and monitoring of persons in the setting of the study

5.4.1 Criteria and methods for early permanent discontinuation of treatment or exclusion of a person from study

The treatment (sildenafil or placebo) may be discontinued by the investigator for any of the following reasons:

- Appearance of adverse treatment event.
- Revascularization

For patients who discontinue the treatment prematurely, the circumstances must be documented in the CRF. These patients will not be excluded from the study (the only reason for exclusion is withdrawal of consent); they will be treated followed in accordance with the intention to treat principle. The IMP supplied to these patients must be retrieved completely and may not be redistributed to other patients.

If a patient discontinues, a complete final examination as scheduled for the final visit should be performed.

5.4.2 Methods of replacement of such person if applicable

This study will be conducted with intention to treat; therefore, patients withdrawn from the study will not be replaced.

5.4.3 Methods of monitoring these persons

Withdrawal of a patient from the study will not in any way change his usual management in relation to his disease. In the case of an adverse event, serious or not, all specific management can be planned depending on severity of the adverse event.

6 TREATMENTS ADMINISTERED TO PERSONS WHO ARE SUBJECTS IN THE STUDY

6.1 Description of the treatment necessary for conduct of the study

6.1.1 Investigational medicinal product(s) / comparator

6.1.1.1 Identification of treatment

♦ IMP1: sildenafil citrate 140 mg

capsule dosed from in 140 mg

The composition is:

sildenafil citrate 140 mg

Excipient: cellulose microcristalline

QSP: 1 capsule N°0 ivory

The route of administration is oral, single morning dose.

The producing laboratory is CHU Brest pharmacy's (PPRIGO)

Sildenafil citrate 140 mg is equal to citrate base 100 mg.

♦ IMP 2: Placebo

It is in the form of a capsule

Excipient: cellulose microcristalline

QSP: 1 capsule N°0 ivory

The route of administration is oral, single morning dose.

The producing laboratory is CHU Brest pharmacy’s (PPRIGO).

6.1.1.2 Packaging and labelling

All blisters will be identically labelled. However, the randomization number will be the only differentiating feature between the different drug packs.

Labels for the delivery: see appendix

6.1.1.3 Manufacturing and shipping

The blinded drugs will be prepared by the PPRIGO hospital pharmacist's consortium (Production Pharmaceutique pour la Recherche Institutionnelle du Grand Ouest) according to GMP conditions.

PPRIGO will provide numbered and labelled blisters each containing 32 capsules of the study drug (sildenafil or placebo according to the randomization order).

6.1.1.4 6.1.1.4. Administration

The administration of experimental drugs sildenafil citrate 140 mg/day or placebo single morning intake is realised by oral route.

6.1.1.5 Dosage adjustment

No adaptation of dosages is required.

6.1.1.6 Special warning and precautions of use

Refer to the summary of product characteristics of Viagra® (sildenafil) (Appendix)

6.1.1.7 Patient Card

According to the Good Manufacturing Practices dated May 26, 2006, a patient card will be given systematically to each patient and marked "Please keep this card with you at all times" specifying the name, address, the name of the study, the patient identification, the starting date of treatment and the treatment number at every visit to the patient.

Typical example of the patient's card: see appendix.

6.2 Methods for observation in the follow-up period

As described in the follow-up parameters analysis, at each visit, patients are asked to even provide the empty blisters of the studied drugs. Similarly, during the phone call, patients will be asked to declare the number of capsules remaining.

The number of capsules remaining at the date of exam is measured.

Compliance will be measured by prospective capsule count.

The initials and the patient's number as well as the date of dispensation are completed on the label of the bottle by the pharmacy at the time of the dispensation of treatments to the patient.

After use, empty, full or partially used blisters are brought back by the patient to every centre. The date of return is completed on the label of the bottle by the pharmacy at the return time of treatments by the patient.

At the end of the treatment, blisters are brought back to the pharmacy at every investigational centre.

6.3 Storage conditions

All products must be stored at room temperature (< 30°C) and protected from light.

6.4 Stocks management, resupplying and dispensation of treatments

PPRIGO (via the pharmacy of the Rennes hospital) will be in charge of the stocks management, resupplying and dispensation of treatments.

A pharmaceutical standard operating procedure will specify all the modalities of supply, storage, dispensation, management of stock and traceability of experimental drugs in each study centre and is passed to each centre with the usual administrative documents (AEC, CPP, and protocol).

6.5 Number of weeks of treatment for the studied drug delivered

The goal is to deliver the required number of capsules + approximately one week to cover the interval allowed to organise the follow-up visits: Thus 32 capsules at the inclusion visit, another 64 capsules at week 4 and 96 capsules at week 12.

6.6 Other treatments: Non-experimental study treatment

Both groups will receive, as standard of care, an Optimal medical treatment including:

Antiplatelet / Direct Oral Anticoagulant + Lipid Lowering Drugs + AT2 Antagonists / ACE Inhibitors (unless contra-indicated) + advice to walk

6.7 Authorised and unauthorised medicinal products and treatments in the setting of the protocol, including emergency medicinal products

6.7.1 Authorised treatments

All except the combination of three or more vasodilator drugs (calcium channel blockers, ACE inhibitors, ARA II, alpha blockers, PDE5 inhibitors) and the combination with sacubitril/valsartan.

The treatments, taken by the patient at the time of inclusion in the study can be continued during the study and can be modified by the referring doctors of the patient without questioning the participation of patients in the study.

Non-study-specific examinations and treatments prescribed during the study period are left to the discretion of the physician in charge of the patient. This protocol makes it possible to envisage etiologies that may be unrecognized but does not, in itself, determine the therapeutic management of study participants. As a result, accident or complication incidents of such treatments are not considered as resulting to the participation in the study: accident or complication incidents of such treatments will be notified as adverse event with etiology “unrelated to the research”.

6.7.2 Unauthorised treatments

Other PDE inhibitors, nitrates, NO donors, poppers, drugs known to interact with PDE5-inhibitors (ritonavir, alpha-blockers…), guanylate cyclase stimulators (riociguat)

6.7.3 Emergency treatment

Male patients will be given a course of action in case of priapism (urgent consultation in Urology).

7 EVALUATION OF SAFETY

7.1 Definitions

- Adverse event (AE): any untoward medical occurrence in a person who is a subject in a biomedical research study, whether or not this occurrence is related to the study or to the studied product.
- Serious Adverse Event (SAE): Severity is defined as one of the following occurrences:
  - Death
  - A life-threatening condition
  - Persistent or significant disability or incapacity
  - Hospitalisation
  - Prolongation of hospitalisation
  - Malformation/congenital anomaly
  - Potentially serious event (an adverse clinical event or a laboratory test result with a serious feature or considered as such by the investigator)
- Adverse Reaction (ADR): any untoward and unintended reaction to an investigational medicinal product, whatever the dose administered.
- Serious adverse reaction (SAR): a serious adverse event with a causal relationship to an investigational medicinal product.
- Unexpected adverse reaction: an adverse reaction whose type, severity, intensity or outcome does not agree with information contained in the summary of product characteristics for an authorised investigational medicinal product, or in case of an unauthorised medicinal product, in the investigator's brochure.
- Causal relationship: relation between the AE and the studied treatment. An Adverse Event related to an investigational medicinal product will become an AR.

7.2 Investigator's role

7.2.1 Notification of serious adverse events

Every serious adverse event related to the studied treatment or not, expected or unexpected, must be reported without delay by the investigator to the sponsor on a "Serious adverse event" form on which will be indicated the date of occurrence, criterion of severity, intensity, relationship with the treatment (or the study) evaluated, and the outcome.

It must be sent by fax to the following number: 02-99-28-40-10 or by mail at vigilance-essais@chu-rennes.fr.

The period in which serious adverse events should be reported begins from the day of the written informed consent up to week 24. The investigator should report **any SAE** that occurs **after these time periods** that is believed to be **related** to study drug.

The narrative must be completed and sent to the sponsor as soon as new relevant information is obtained. Copies of the patient's medical record must be attached as well as results of laboratory tests.

Whenever a serious adverse event persists at the end of the study, the investigator must follow the patient until the event is considered resolved.

7.2.2 Notification of non-serious adverse events

All other Adverse Events reported from the day of the written informed consent up to week 24, will be reported on the "adverse event" form of the case report form specifying:

- Description
- Date of occurrence
- Date of end
- Severity
- Measures with regard to experimental treatment
- Causality

7.2.3 Adverse events not to be reported

Following hospitalisations will not be notified to the sponsor:

- Hospitalisation for social or administrative reason
- Hospitalisation scheduled for the study
- Admission in day hospital
- Hospitalisation for usual treatment or observation of the disease under study without deterioration of the patient
- Hospitalisation for medical treatment or surgery scheduled before the inclusion of the patient
- Hospitalization for surgery (revascularization treatments)

7.2.4 Medication errors

The definition of suspected unexpected serious adverse reaction (SUSAR) covers also medication errors and uses outside what is foreseen in the protocol, including misuse and abuse of the product.

7.2.5 Exposure during pregnancy

If, following initiation of sildenafil/placebo, it is subsequently discovered that a patient is pregnant or may have been pregnant at the time of sildenafil/placebo exposure, including during at least 6 half-lives after study drug administration, the sildenafil/placebo will be permanently discontinued.

From the day of the written informed consent up to one week after the last administration of the experimental treatment, the investigator must immediately notify the sponsor of this event via the pregnancy initial surveillance form.

Follow-up information regarding the course of the pregnancy, including perinatal and neonatal outcome must be reported on the pregnancy outcome surveillance form.

7.3 Sponsor's role

7.3.1 Analysis of serious adverse events

The sponsor must evaluate the following:

- the causal relationship of serious adverse events (all adverse events for which the investigator or the sponsor considers that a causal relationship with the investigational medicinal products can be reasonably considered, are considered as suspected adverse reactions. If the sponsor's evaluation of the event differs from the investigator's, both opinions are mentioned in the statement sent to the competent authority if this statement is necessary),
- and their expected or unexpected feature, using the reference document in force.

7.3.2 Scoring of causal relationship

The method used by the sponsor in evaluating the relationship of the event is as follows (EVCTM method):

- Unrelated: the event occurred within a time period not compatible compared to administration of the medicinal product, and/or a sufficient amount of information exist showing that the observed reaction is unrelated to the medicinal product, and/or a probable alternative explanation exists.
- Possible: the event occurred within a period compatible after administration of the medicinal product and although the responsibility of the product cannot be ruled out, other factors can be implicated, such as the patient's clinical condition or concomitant

administration of other medicinal products. Information on the outcome upon discontinuation of the medicinal product can be absent or inconclusive.

Adverse events whose relationship with the investigational medicinal products is possible are considered as related to the product(s). If they are unexpected, they are qualified as being Suspected Unexpected SAR (SUSAR) and must be notified in a report by the sponsor (see Following paragraph).

7.3.3 Declaration of unexpected serious adverse events

The sponsor reports all suspected unexpected serious adverse reactions (SUSAR) to Eudravigilance (European pharmacovigilance database) and to local regulatory agency when applicable within the regulatory time periods for reporting:

- Without delay when the gravity is “death” or “life threatening”
- Within 15 days for the other types of gravity.

7.3.4 Transmission of annual safety reports

On the anniversary date of authorisation of the study delivered by the Health Authorities, the sponsor writes a safety report containing:

- the list of serious adverse events which may be related to the investigational medicinal product(s) studied, including expected and unexpected serious effects,
- a succinct and critical analysis of safety of patients who are subjects in the study.

It is sent to local regulatory agency and to the ethics committee within 60 days following the anniversary of authorisation of the trial.

7.3.5 Declaration of other safety data

Every safety finding that can alter the benefit/risk ratio of an investigational medicinal product or sufficient to consider changes in administration of the study product or changes in conduct of the trial must immediately be notified to the sponsor. The former will be the subject of a report to the competent authorities without delay.

7.4 Expected adverse effects

The reference safety information is the SmPC of Viagra® (sildenafil). Expected adverse effects are listed in this document.

8 STATISTICS

Statistical analysis will be performed on all randomized and evaluated patients and according to the treatment arm (intention to treat analysis). It will be performed with SAS software (SAS Institute, Cary, NC, USA) in the Biometrics Unit department of pharmacology of Rennes. Specific details will be provided in the Statistical Analysis Plan.

8.1 Description of planned statistical methods, including schedule of planned interim analysis

8.1.1 Comparison of groups at inclusion

Demographic and baseline characteristics will be summarised overall and by treatment. For continuous variables, the summary will consist of descriptive statistics (number, mean, standard deviation, minimum, median, and maximum). For categorical variables, the summary will consist of number and percentage of subjects in each category.

8.1.2 Analysis of the primary criteria

The analysis will be performed when the number of subjects required (220 patients) will have been reached. The primary endpoint is the absolute change of the ACD from baseline to week 24. Analysis for this endpoint will be performed using Student's t-test or the Wilcoxon test as appropriate.

8.1.3 Analysis of other criteria

For *quantitative endpoints*, analysis will be performed using Student's t-test or the Wilcoxon test as appropriate, with covariate adjustment performed by exact linear regression if substantial baseline imbalances are detected for any important prognostic factors.

For *binary endpoints*, analysis will be performed using the chi-square test, with covariate adjustment performed by exact logistic regression if substantial baseline imbalances are detected for any important prognostic factors.

For *time-to-event endpoints*, Kaplan-Meier estimates of the survival functions will be presented. Treatment comparison will be made via a log-rank test. Cox's regression will be performed to obtain the hazard ratio estimate and the corresponding 95% CI. Additional Cox's regression may be performed to include appropriate baseline prognostic variables and treatment group.

For endpoint evaluated several times during the monitoring will be compared with mixed model.

8.1.4 Analysis of adverse events

Safety will be evaluated for all subjects who have received at least 1 dose of study drug. Adverse events (AEs) and serious adverse events (SAEs) will be coded using the current version of Medical Dictionary for Regulatory Activities (MedDRA).

8.1.5 Planned number of persons to be enrolled in the study, and planned number of persons in each centre with its statistical justification

To detect an effect size of at least 0.5 for sildenafil compared with placebo, with power of 95% and significance level of 0.05 (two-sided) with equal allocation to treatment groups, 110 randomized patients will be required for each treatment group.

Effect-size	0.5	[Sildenafil: 110 patients] [Placebo: 110 patients] [Total: 220 patients]
Ratio	1:1	
Formulation	Two-sided	
α	5%	
Power	95%	
Software	R [power.t.test]	

8.2 Planned degree of statistical significance

A p-value <0.05 will be considered as significant for all analyses.

8.3 Method of management of missing, unused or invalid data

Missing data will not be replaced.

8.4 Management of changes made to plan of analysis of initial strategy

Any deviations from the planned analyses will be described and justified in the final integrated study report.

8.5 Choice of persons to include in the analysis

This trial is an intention to treat study that is all randomized and evaluated patients will be analysed in their randomization group.

9 RIGHT OF ACCESS TO DATA AND SOURCE DOCUMENTS

9.1 Access to data

In compliance with GCP:

- the sponsor is in charge of obtaining the agreement of all parties involved in the study to ensure direct access to all sites of study conduct, source data, source documents and reports with the aim of control quality and of an audit by the sponsor,
- investigators will make available to persons in charge of monitoring, control quality or audit of the study documents and individual data which are strictly necessary for such checking, in compliance with legislative and regulatory conditions in force (articles L.1121-3 and R.5121-13 of the French public health code).

9.2 Source documents

Source data defined as all documents or original items which make it possible to demonstrate the existence or accuracy of data or of a finding recorded in the clinical study will be kept for 25 years by the investigator or by the hospital if it involves a hospital medical dossier.

Source documents consist of a medical dossier, originals of laboratory test results, imaging examination reports, etc. Other source document specially created for the study can be used if they are dates and signed by the investigator.

9.3 Confidentiality of data

In compliance with conditions concerning confidentiality of data to which persons in charge of quality control of the biomedical study have access (article L.1121-3 of the French public health code and *General Data Protection Regulation*), in compliance with conditions pertaining to confidentiality of information in particular concerning the type of investigational medicinal products, the tests, persons who are subjects in the study and results obtained (article R. 5121-13 of the French public health code), persons having direct access will take all necessary precautions with the aim of ensuring confidentiality of information pertaining to the investigational medicinal products, tests, the persons who are subjects in the study and in particular concerning their identity as well as that of results obtained.

These persons, in the same capacity as the investigators themselves, are subject to professional secrecy (according to conditions defined by articles 226-13 and 226-14 of the French penal code).

During the study or at its end, data collected on persons who are subjects in the study and forwarded to the sponsor by the investigators (or all other specialised participants) will be made anonymous (deletion of subjects' names).

They must not in any event clearly reveal the names of the persons concerned or their address.

Only the first letter of the surname and of the first name of the subject are recorded, together with the code number specific for the study indicating the order of inclusion of subjects.
The sponsor will make certain that each person who is a subject in the study has provided his or her written agreement for access to his/her personal individual data and strictly necessary for quality control of the study.

10 QUALITY CONTROL AND ASSURANCE

A Clinical Research Associate (CRA) designated by the sponsor will ensure proper conduct of the study and the quality of data collected, in agreement with the Standard Operating Procedures applied in the CHU of Rennes and in compliance with Good Clinical Practice as well as legislative and regulatory conditions in force.
The investigator and members of his team agree to make themselves available at Quality Control visits carried out at regular intervals by the Clinical Research Associate. At these visits the following items will be reviewed:

- informed consent,
- compliance with study protocol and procedures defined in it,
- quality of data collected in the case report forms: accuracy, missing data, consistency of data with source documents (medical dossiers, appointment diaries, originals of laboratory test results, etc.),
- management of possible products.

In addition, the investigators agree to receive quality assurance audits performed by the sponsor as well as inspections performed by the Competent Authorities. All data, all documents and reports can be subjected to audits and regulatory inspections without being opposed by medical secrecy.

11 NOTIFICATION OF SERIOUS BREACHES AND DEVIATIONS

Deviations from clinical trial protocols and GCP may occur in clinical trials. The majority of these instances are technical deviations that do not result in harm to the trial subjects or significantly affect the scientific value of the reported results of the trial. These cases should be documented (for example, in the trial case report form or the trial master file) in order for appropriate corrective and preventative actions to be taken.
Serious breaches are deviations likely to affect to a significant degree the safety and rights of a subject or the reliability and robustness of the data generated in the clinical trial.

The sponsor shall notify the Member States concerned about a confirmed serious breach according to the procedure in force of the EU regulation n°536/2014 or of the version of the protocol applicable at the time of the breach through the EU portal CTIS without undue delay but not later than seven days of becoming aware of that breach.

12 ETHICAL CONSIDERATIONS

12.1 Regulatory and institutional review

The protocol, information form and certificate of consent of the study will be submitted to the relevant independent ethics committee (IEC) for review.

Notification of a favourable opinion from the IEC will be transmitted to the sponsor of the study and to the competent Authority. An application for authorisation will be sent by the sponsor to local regulatory authority prior to start of study.

12.2 Substantial changes

In case of a substantial change made to the study protocol by the investigator, it will be approved by the sponsor. The latter must obtain prior to start of study a favourable opinion from the ethics committee and authorisation from the relevant health authority in the setting of their respective competence. Additional consent from persons participating in the study will be collected if necessary.

12.3 Information for patients and written informed consent form

Patients will be completely and truly informed in terms which are understandable to them of the objectives and constraints of the study, the possible risks incurred, measures of supervision and safety necessary, of their right to refuse to participate in the study or of the possibility of retracting their agreement at any time.

All this information is contained in an information and consent form given to the patient. Free, informed and written consent from the patient will be collected by the investigator, or a doctor representing him prior to final inclusion in the study. A copy of the information and consent form signed by both parties will be given to the patient; the investigator will keep the original of it.

12.4 Compensation of subjects

No compensation for subjects is planned by the protocol.

12.5 Registration in the national file of persons who are subjects in biomedical research

Such registration is not planned as part of the research.

13 DATA PROCESSING AND RETENTION OF DOCUMENTS AND DATA

13.1 Case report forms and data entry

All information required by the study protocol must be recorded in the case report forms and an explanation must be provided for any missing data. Data must be collected progressively as obtained and recorded explicitly in the case report forms.

An electronic case report form (e-CRF) will be made available and data entry will be completed in centres through a WEB interface (Ennov Clinical software, Ennov, Paris (75), France). It solely requires an internet connection and a browser. An aid document for use of this tool will be provided to investigators. The interface between the CRA and the investigator thus will be promoted, making possible the collection and control of data at a distance. Tests of control of consistency of data will be incorporated in electronic format. An audit function is incorporated in the e-CRF thus making it possible to follow any change in study data. This function also makes it possible to clearly identify the person who made a change as well as the date. A justification possibly can be incorporated in comment.

If requested, a paper copy will be printed at the end of the study, authenticated (dated and signed) by the investigator and a copy will be sent to the sponsor and archived.

Data analysis will be carried out by statisticians of the Biostatistics Unit Department of Pharmacology of Rennes.

13.2 CNIL

This study falls within the scope of the “Methodology of Reference” (MR-001) in application of conditions of article 54 line 5 of French law no. 78-17 of 6 January 1978 modified pertaining to data processing, computer files and liberties. This change has been authorised by the decision of 5 January 2006. The CHU of Rennes, sponsor of the study, has signed an agreement of compliance with this “Methodology of Reference”.

13.3 Archiving

The following documents will be kept in the respective departments until the end of the period of practical utility.

These documents are:

- The study protocol and its annexes, and possible amendments,
- The original signed informed information and consent forms,
- Individual data (authenticated copies of raw data),
- Monitoring documents,
- Statistical analyses,
- The study final report.

At the end of the period of practical utility, the documents will be archived by the sponsor for at least 25 years after the end of the study or its early discontinuation, in compliance with institutional practices.

Archived documents cannot be moved or destroyed without agreement of the sponsor. All data, all documents and reports can be subjected to an audit or inspection.

14 INSURANCE

The Sponsor will take out, for the duration of the study, insurance covering the sponsors civil responsibility as well as that of all doctors involved in conduct of the study. They will also insure the total compensation of all harmful consequences of the study for persons who are subjects in it and their heirs, except when evidence is provided that the harm is not causally related to a fault of the sponsor or that of any participant, without it being possible to oppose the intervention of a third party or the voluntary withdrawal of a person who had initially consented to be a subject in the study.

15 STUDY FEASIBILITY

Our department performs over 200 treadmill tests in patients with PAD and arterial claudication patients per year. It is estimated that 1/3 of these patients will not be eligible for the VIRTUOSE study. If one out of two patients accepts to participate in this study, recruitment should reach 65 patients per year (130 patients over two years).

16 RULES PERTAINING TO PUBLICATION

Communications and scientific reports corresponding to this study will be carried out under the responsibility of Dr Loukman OMARJEE, the coordinator investigator of the study. Co-authors of the report and publications will be the investigators and clinicians involved, in proportion to their contribution to the study, as well as the biostatistician and associated researchers.

Dr Loukman OMARJEE, the coordinator investigator of the study will always be first or last author and he is the only one able to validate the list of co-authors.

Unless specifically agreed to by the coordinating investigator, none of the study components should be presented orally or posteriorly before acceptance of the corresponding manuscript in a referenced journal. In addition, the presentation of the results is always made in priority in Anglo-Saxon journals left to the choice of the coordinating investigator on proposal of the Steering Committee.

As recommended by the Vancouver group, the author rating is based on a substantial contribution:

- Conception of the study protocol or analysis and interpretation of the data
- Manuscript writing or critical analysis on important intellectual content
- Final acceptance of the version to be published.

Rules on publication will follow international recommendations (N Engl J Med, 1997; 336:309-315).

The study will be recorded on a freely accessible website (Clinical Trials) prior to inclusion of the first patient in the study.

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18 LIST OF APPENDIX

Study abstract (English and French)

Study team

List of principal investigators

Independent Data and Safety Monitoring Board (DSMB)

Label for delivery

Patient's card

Questionnaire SF-36

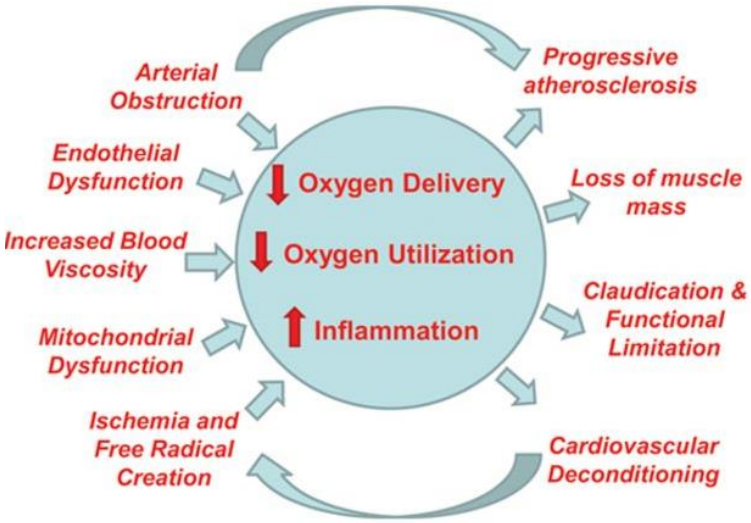
Peripheral Artery Questionnaire

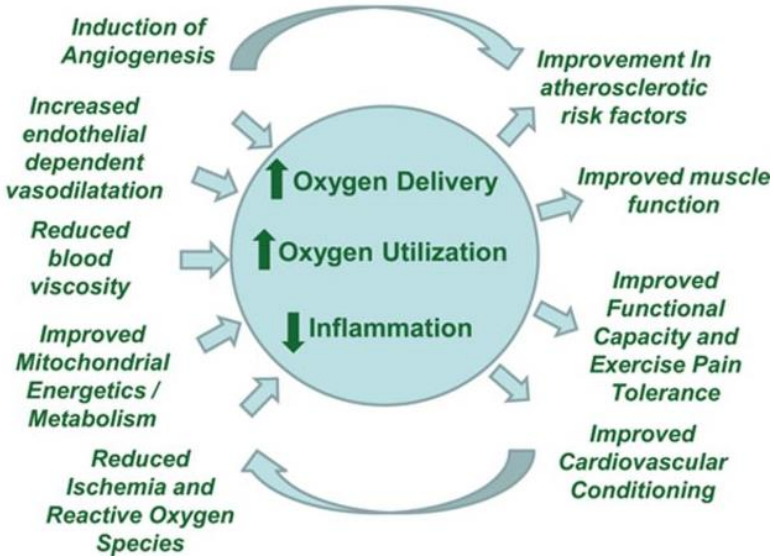
The vicious circle mechanisms circle of Peripheral Artery Disease as proposed by *Bonaca et al*

Typical examples of TcpO2 results

Summary of the drug characteristics

The vicious and virtuous circle mechanisms circle of Peripheral Artery Disease as proposed by *Bonaca et al*.[11]

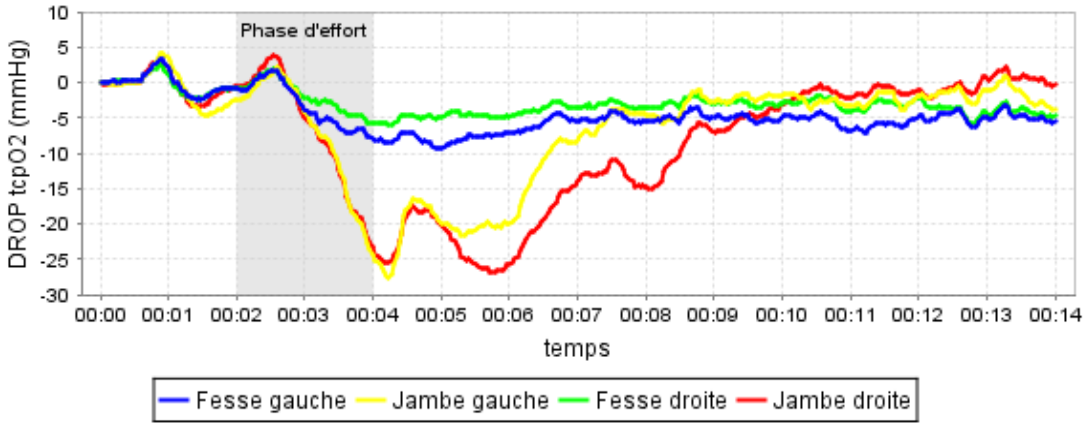




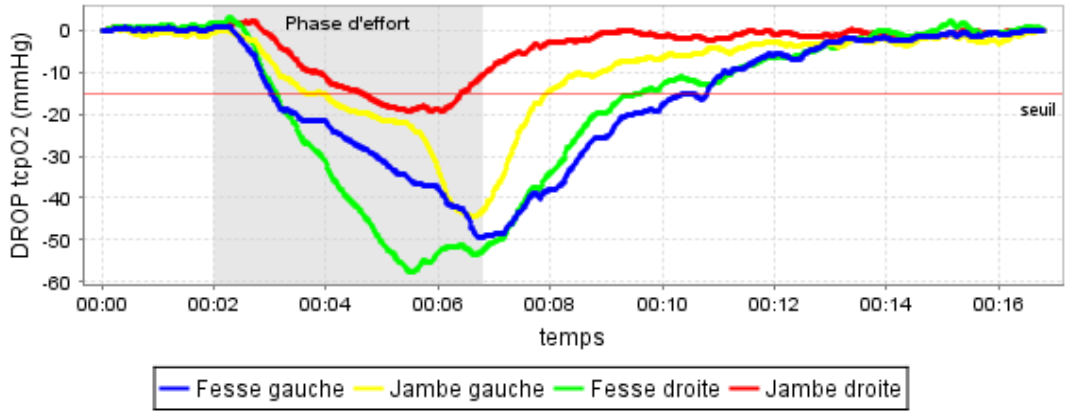
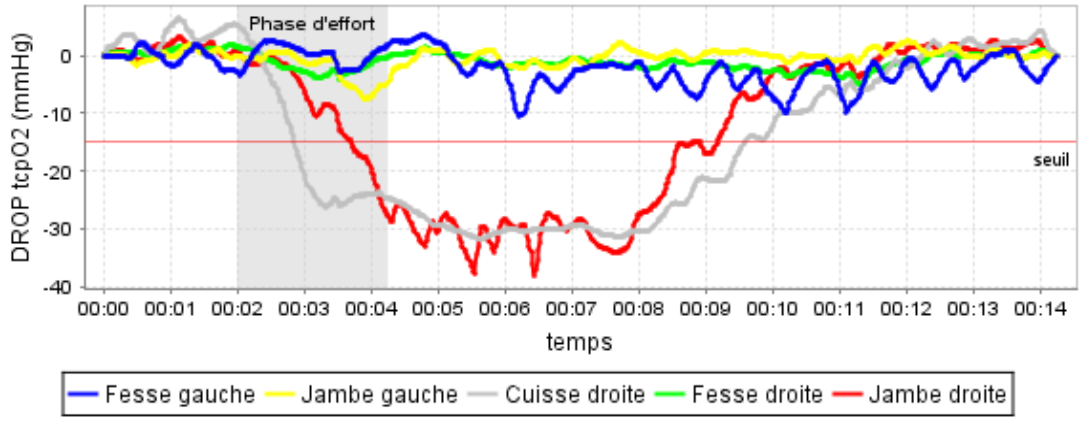
Typical examples of TcpO2 results

As shown the test always starts with a 2-min resting period. The walking period is the Grey square. As shown the test is able to discriminate right to left or proximal from distal ischemia.

Example of bilateral calf ischemia from bilateral femoral occlusion



Example of bilateral proximo-distal ischemia:



Example of unilateral thigh and calf ischemia resulting from external iliac artery occlusion